



Nawatech

Beyond Engineering

nawatech.co



KAI Workshop – Day 6

Prepared for: PT Kereta Api
Indonesia



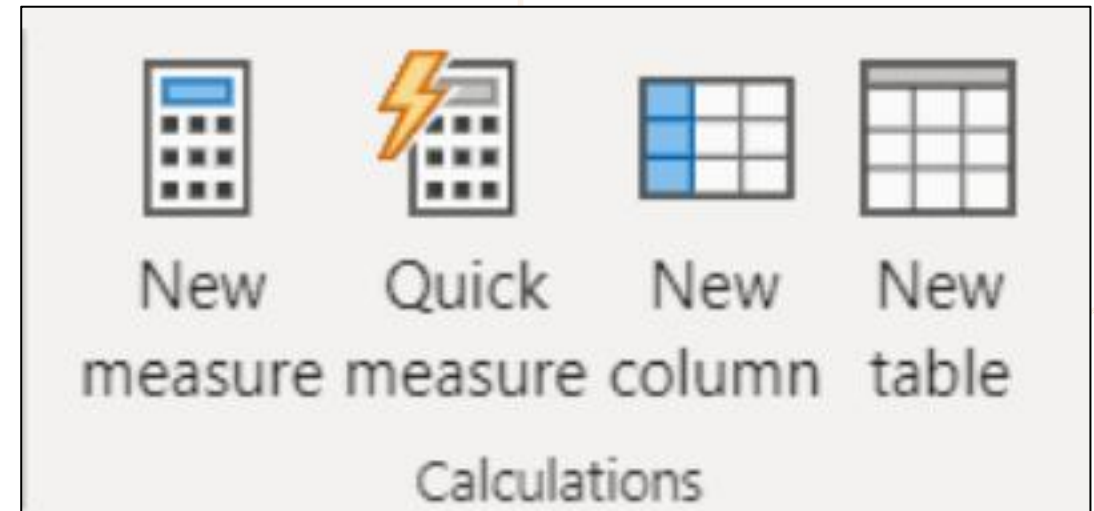
Add measures to semantic
models





What is DAX?

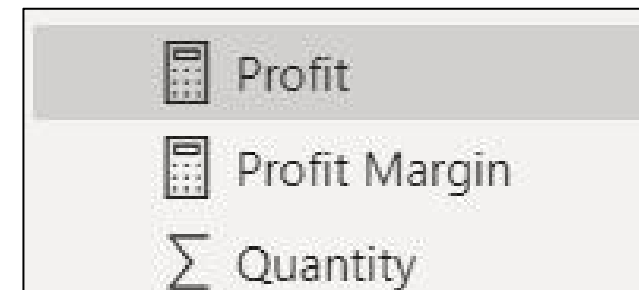
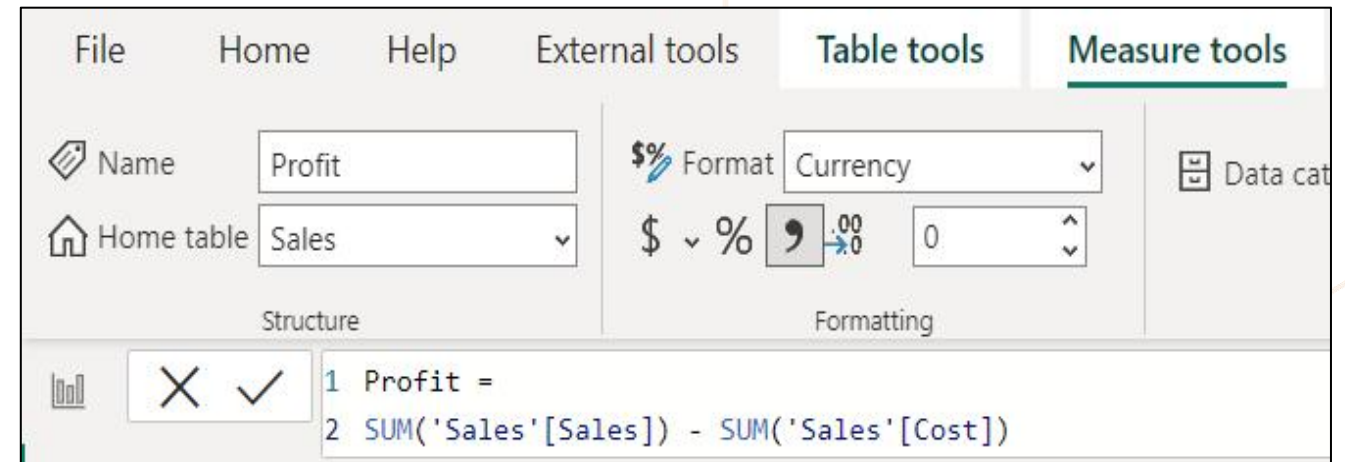
- Data Analysis Expressions
- Library of functions and operators
- Build formulas and expressions
- Create calculated measures, columns, and tables





Create calculated measures

- Defined with DAX definitions
- Computed on the fly.
- Not stored in data model.
- Responsive to interactions.
- Indicated by calculator icon.





Implicit vs. Explicit vs. Quick measures

☐  Sales % All Region

☐  Sales % Country

☐  Sales % Group

☐ Σ Sales

☐ ShipDate

Quick measure

Calculation

Filtered value 

Calculate a value with a filter applied. [Learn more](#)

Base value ⓘ

Add data fields here

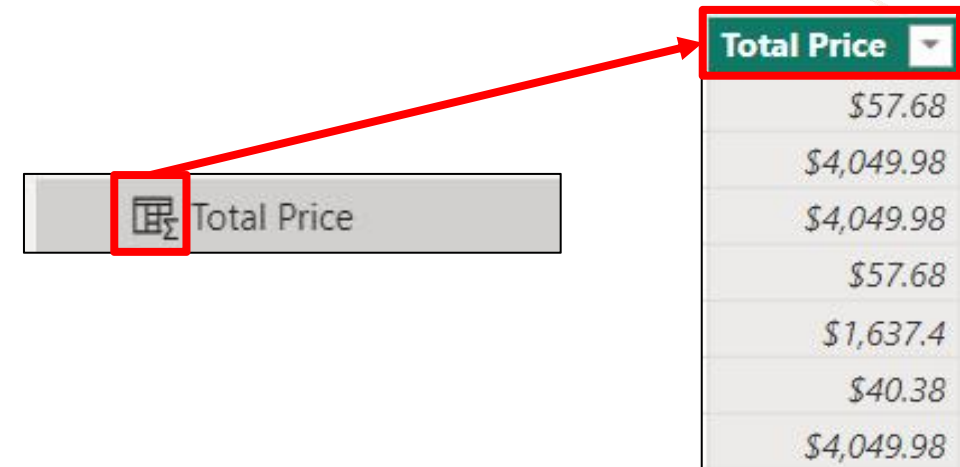
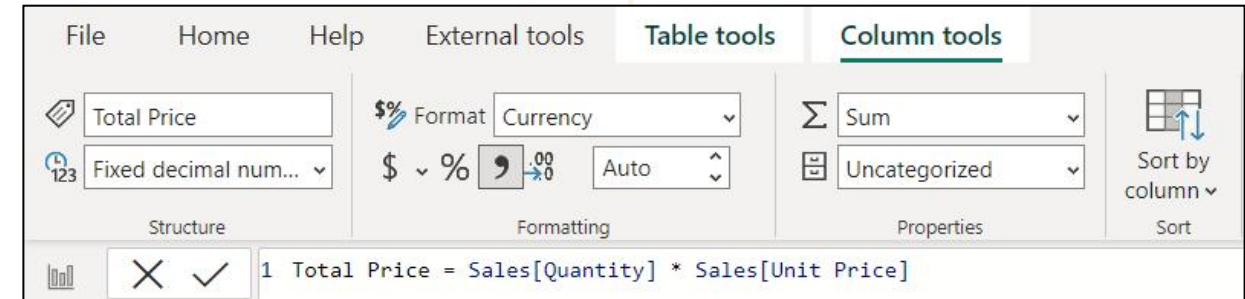
Filter ⓘ

Add data fields here



Create calculated columns

- Defined using DAX expressions.
- Computed & stored in data model.
- Useful “helper/connector columns.”
- Recalculated during data refresh.
- Table and Sigma icon.



Total Price
\$57.68
\$4,049.98
\$4,049.98
\$57.68
\$1,637.4
\$40.38
\$4,049.98



Create calculated tables

- Defined using DAX expressions.
- Computed & stored in data model.
- Useful for aggregating data or creating custom tables.
- Table and calculator icon.

The screenshot shows the Power BI interface with the 'Table tools' ribbon selected. The 'Name' field is set to 'Date'. The 'Structure' section shows the formula bar with the DAX expression: `Date = CALENDARAUTO(6)`. The 'Calculations' section shows icons for 'New measure', 'Quick measure', 'New column', and 'New table'. The 'Data' pane on the right shows the 'Date' table selected, with a list of columns: Date, Year, Quarter, Month, and MonthKey. The main view displays a table with 10 rows of data for the month of July 2017.

Date	Year	Quarter	Month	MonthKey
7/1/2017 12:00:00 AM	FY2018	FY2018 Q1	2017 Jul	201707
7/2/2017 12:00:00 AM	FY2018	FY2018 Q1	2017 Jul	201707
7/3/2017 12:00:00 AM	FY2018	FY2018 Q1	2017 Jul	201707
7/4/2017 12:00:00 AM	FY2018	FY2018 Q1	2017 Jul	201707
7/5/2017 12:00:00 AM	FY2018	FY2018 Q1	2017 Jul	201707
7/6/2017 12:00:00 AM	FY2018	FY2018 Q1	2017 Jul	201707
7/7/2017 12:00:00 AM	FY2018	FY2018 Q1	2017 Jul	201707
7/8/2017 12:00:00 AM	FY2018	FY2018 Q1	2017 Jul	201707
7/9/2017 12:00:00 AM	FY2018	FY2018 Q1	2017 Jul	201707
7/10/2017 12:00:00 AM	FY2018	FY2018 Q1	2017 Jul	201707



Columns vs. measures

- Calculated columns:
 - Create values for each row in table.
 - Store values in the .pbix file.
 - Increases data model size.
 - Row-by-row calculation can impact performance.
 - Must be referenced with measures for reuse.
- Measures:
 - Calculate on demand.
 - Calculated based on filters.
 - Doesn't affect data model size.
 - DAX expressions may still be suboptimal.
 - Can reference other measures directly for reuse.

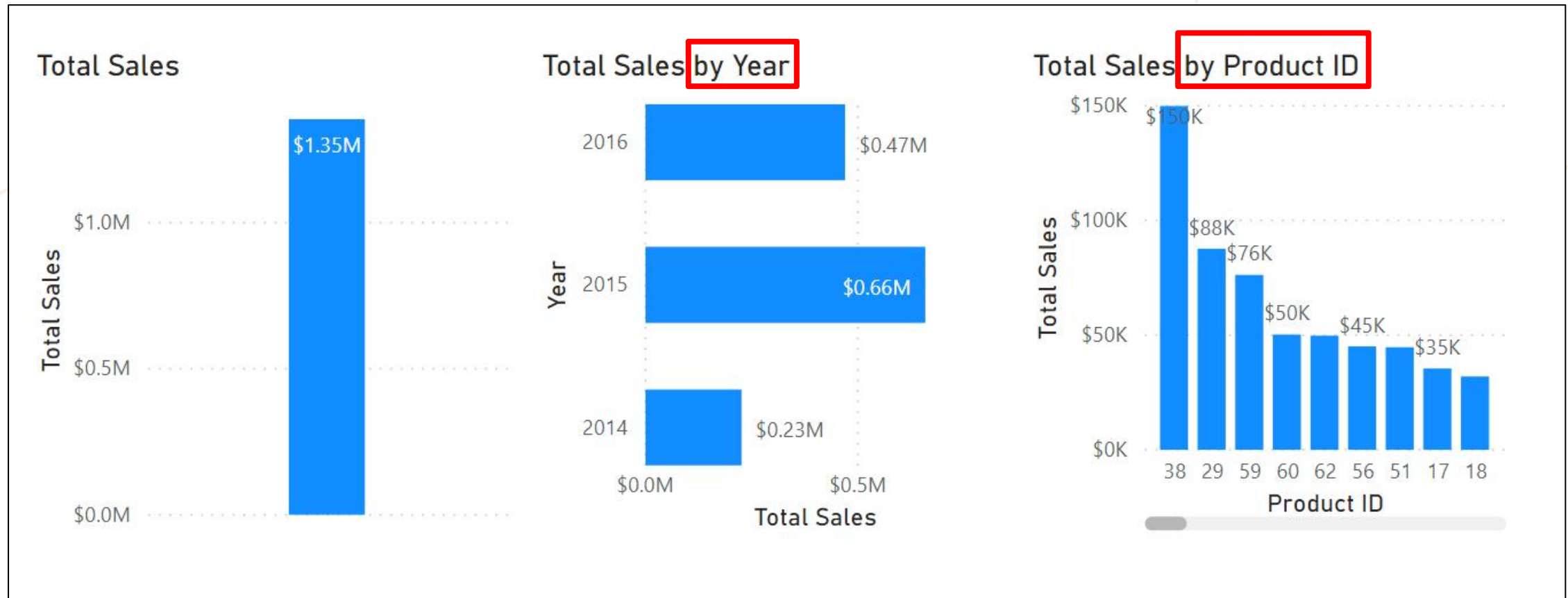
Use DAX functions





Understanding filter context

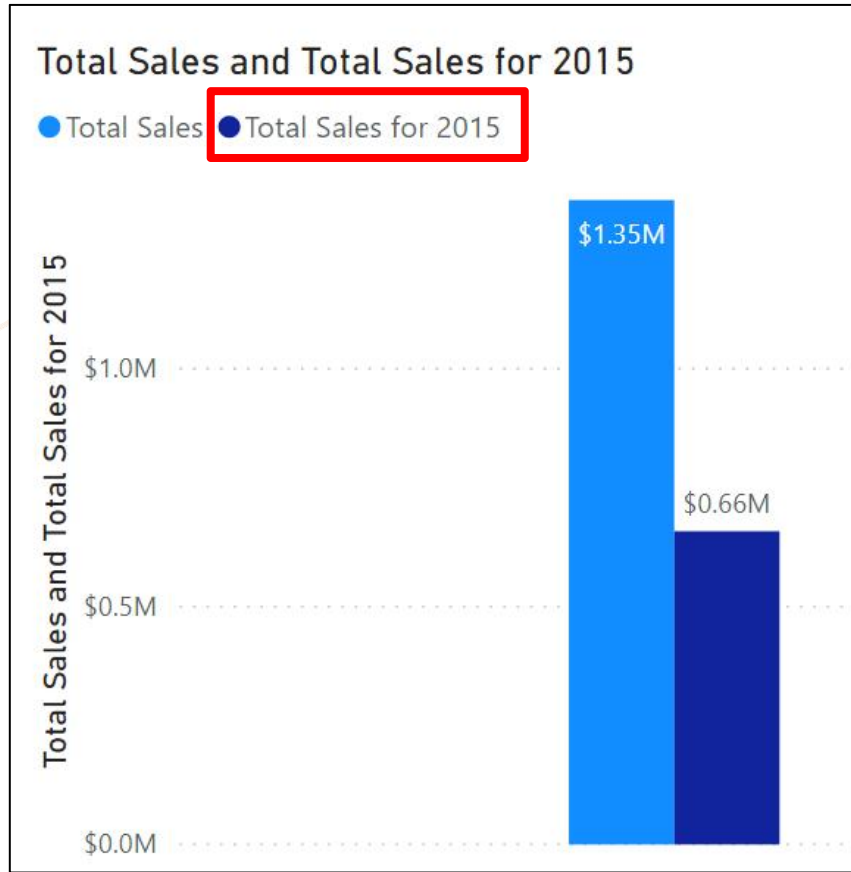
- Measures are contextually different, or “dynamic,” depending on filters.





CALCULATE() function

- Adjusts how measures interpret data filters, enabling context control.



```
Total Sales for 2015 =  
CALCULATE (  
    SUM ( 'Sales OrderDetails'[Total  
Price] ),  
    YEAR ( 'Sales OrderDetails'[OrderDate] )  
= 2015  
)
```

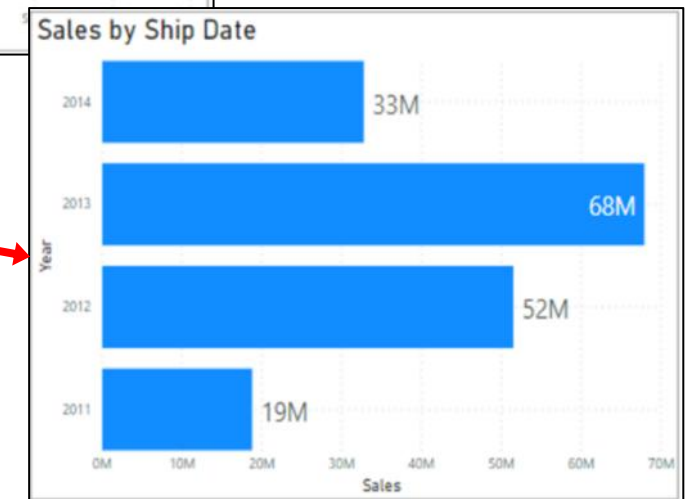


Use inactive relationships with DAX

- Enable additional table filtering without impacting the active relationship.



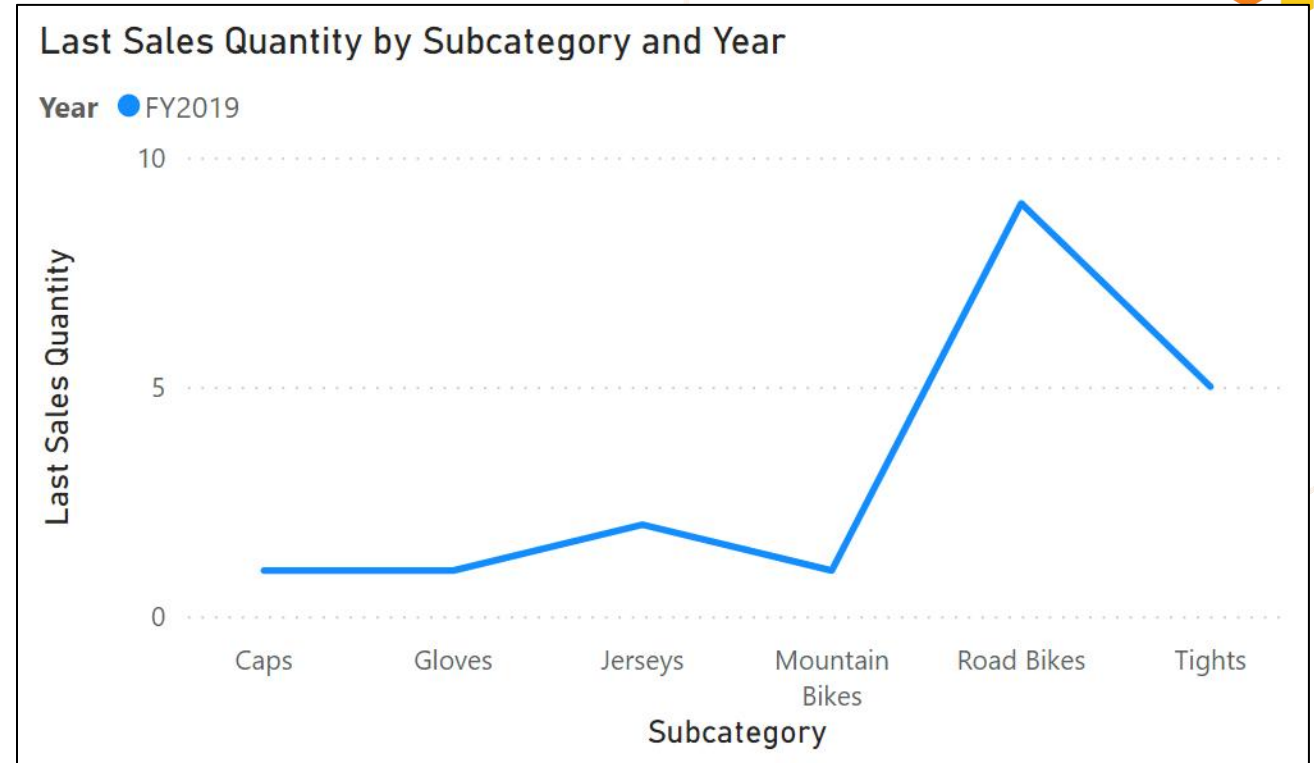
```
Sales by Ship Date =  
CALCULATE (  
    Sales[TotalPrice],  
    USERELATIONSHIP (  
        'Calendar'[Date], Sales[ShipDate])  
)
```





Semi-additive measures

- **Context-dependent:**
Results vary based on dimensions.
- **Time Intelligence:**
Useful for averages, ratios, balances.
- **Caution with aggregation:**
Beware of misleading results when aggregating.



```
Last Sales Quantity =  
CALCULATE(  
    SUM(Sales[Quantity]),  
    LASTDATE('Date'[Date])  
)
```



Create a common date table

Mark as date table

Select a column to be used for the date. The column must be of the data type 'date' and must contain only unique values. [Learn more](#)

Date column

Date

✓ Validated successfully

When you mark this as a date table, the built-in date tables that were associated with this table are removed. Visuals or DAX expressions referring to them may break. [Learn how to fix visuals and DAX expressions](#)

OKCancel

✓ Date

☐ Date

✓ ☐ Fiscal

☐ Year

☐ Quarter

☐ Month

☐ Month

☐ Quarter

☐ Year



Time intelligence functions

Month	2014	2015	2016
January		\$66,692.8	\$100,854.72
February		\$107,900	\$205,416.67
March		\$147,879.9	\$315,242.12
April		\$203,579.29	\$449,872.68
May		\$260,402.99	\$469,771.34
June		\$299,490.99	\$469,771.34
July	\$30,192.1	\$354,955.92	\$469,771.34
August	\$56,801.5	\$404,937.61	\$469,771.34
September	\$84,437.5	\$464,670.63	\$469,771.34
October	\$125,641.1	\$534,999.13	\$469,771.34
November	\$175,345.1	\$580,912.49	\$469,771.34
December	\$226,298.5	\$658,388.75	\$469,771.34
Total	\$226,298.5	\$658,388.75	\$469,771.34

Total Sales Previous Month =

```
CALCULATE ( SUM ( Sales[Total Price] ),  
PREVIOUSMONTH ( 'Date'[Date] ) )
```

Year	Month	Total Sales	Total Sales Previous Month
2015	March	\$39,979.9	\$41,207.2
2015	April	\$55,699.39	\$39,979.9
2015	May	\$56,823.7	\$55,699.39
2015	June	\$39,088	\$56,823.7
2015	July	\$55,464.93	\$39,088
2015	August	\$49,981.69	\$55,464.93
2015	September	\$59,733.02	\$49,981.69
2015	October	\$70,328.5	\$59,733.02
2015	November	\$45,913.36	\$70,328.5
2015	December	\$77,476.26	\$45,913.36



Use variables to improve performance and readability

Without variable:

Sales YoY Growth =

```
DIVIDE (  
    ( [Sales] - CALCULATE ( [Sales], PARALLELPERIOD ( 'Date'[Date], -12, MONTH ) ) ),  
    CALCULATE ( [Sales], PARALLELPERIOD ( 'Date'[Date], -12, MONTH ) ) )
```

With variable:

Sales YoY Growth =

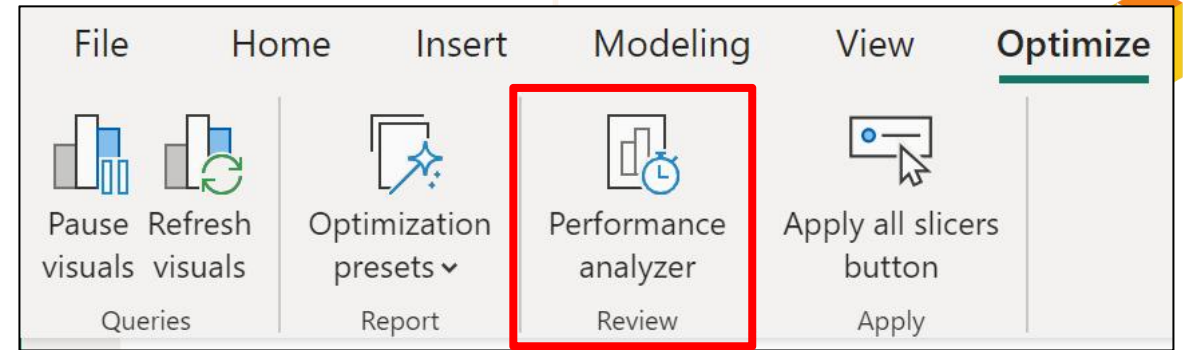
```
VAR SalesPriorYear = CALCULATE ( [Sales], PARALLELPERIOD ( 'Date'[Date], -12, MONTH ) )  
VAR SalesVariance = DIVIDE ( ( [Sales] - SalesPriorYear ), SalesPriorYear )  
  
RETURN  
SalesVariance
```

Optimize model performance



Tune report performance

- Performance analyzer in Power BI Desktop
- Record report interactions.
- Review query and rendering times.
- Identify bottlenecks, optimize queries.



Performance analyzer

Start recording Refresh visuals Stop

Clear Export

Name	Duration (ms)
Recording started (6/19/2023 3:12:20 PM)	-
Cross-highlighted	-
Button	92
Image	91
Sales and Profit Margin by Month	359
DAX query	19
Visual display	28
Other	312
Copy query	
Quantity by Category	302
Slicer	129
Slicer	222



Analyze query plans for optimization opportunities

```
Count Customers =  
CALCULATE (  
    DISTINCTCOUNT ( Order[ProductID] ),  
    FILTER ( Order, Order[OrderQty] >= 5 )  
)
```

```
Count Customers =  
CALCULATE (  
    DISTINCTCOUNT ( Order[ProductID] ),  
    KEEPFILTERS (Order[OrderQty] >= 5 )  
)
```

☐ Sales by Year	
DAX query	2754
Visual display	57
Other	160
Copy query	

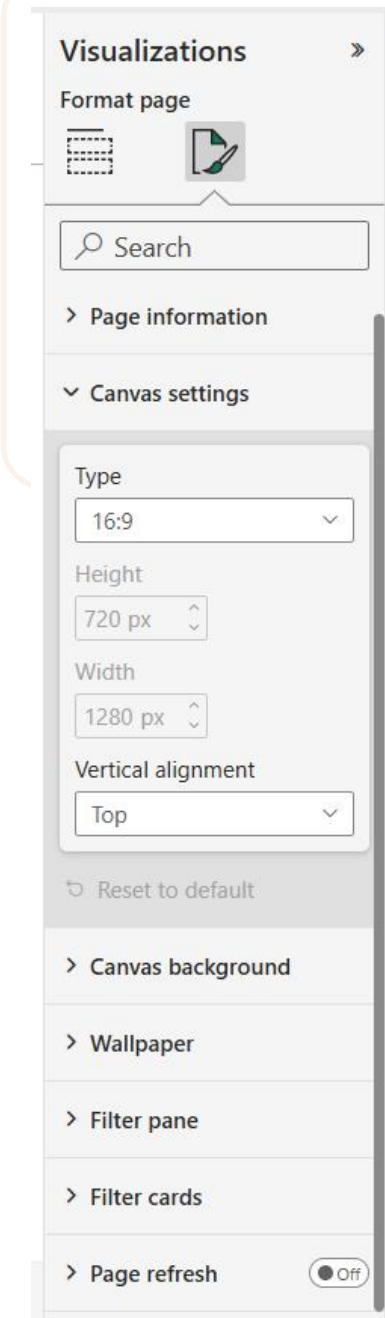
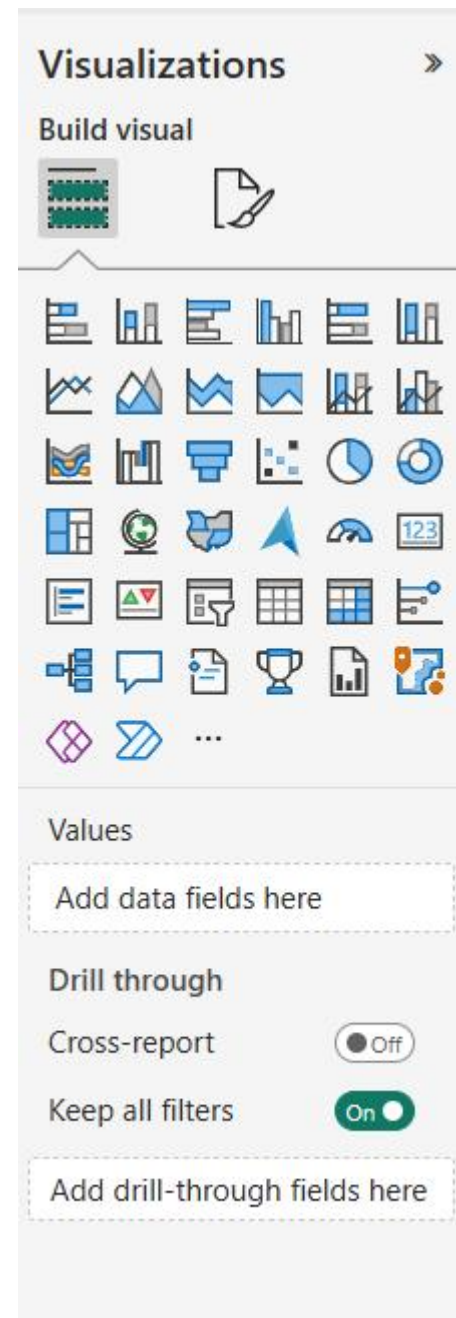
☐ Sales by Year	
DAX query	54
Visual display	57
Other	160
Copy query	

Create visual calculations



Create Visualization

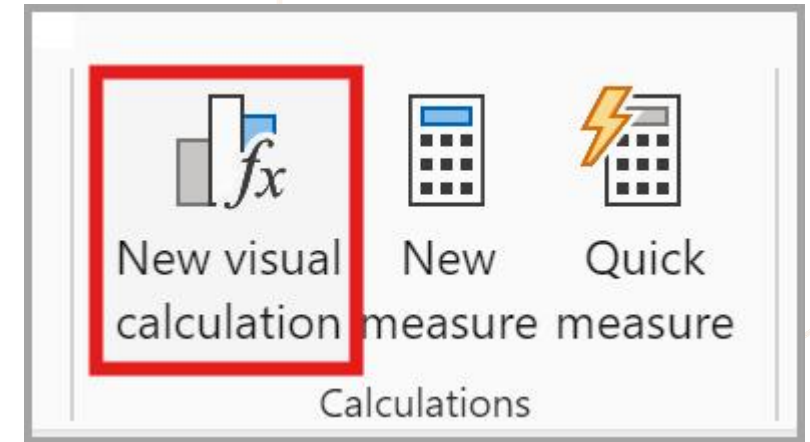
- Visualization
 - Build : To create various chart or graph
 - Format : To customize the chart or graph





What are visual calculations?

- DAX calculations
- Stored on the visual
- Visual representation of calculations



Compared to measures that are stored in the semantic model and must be added to a visual.



Explore the visual calculation interface





Create a visual calculation

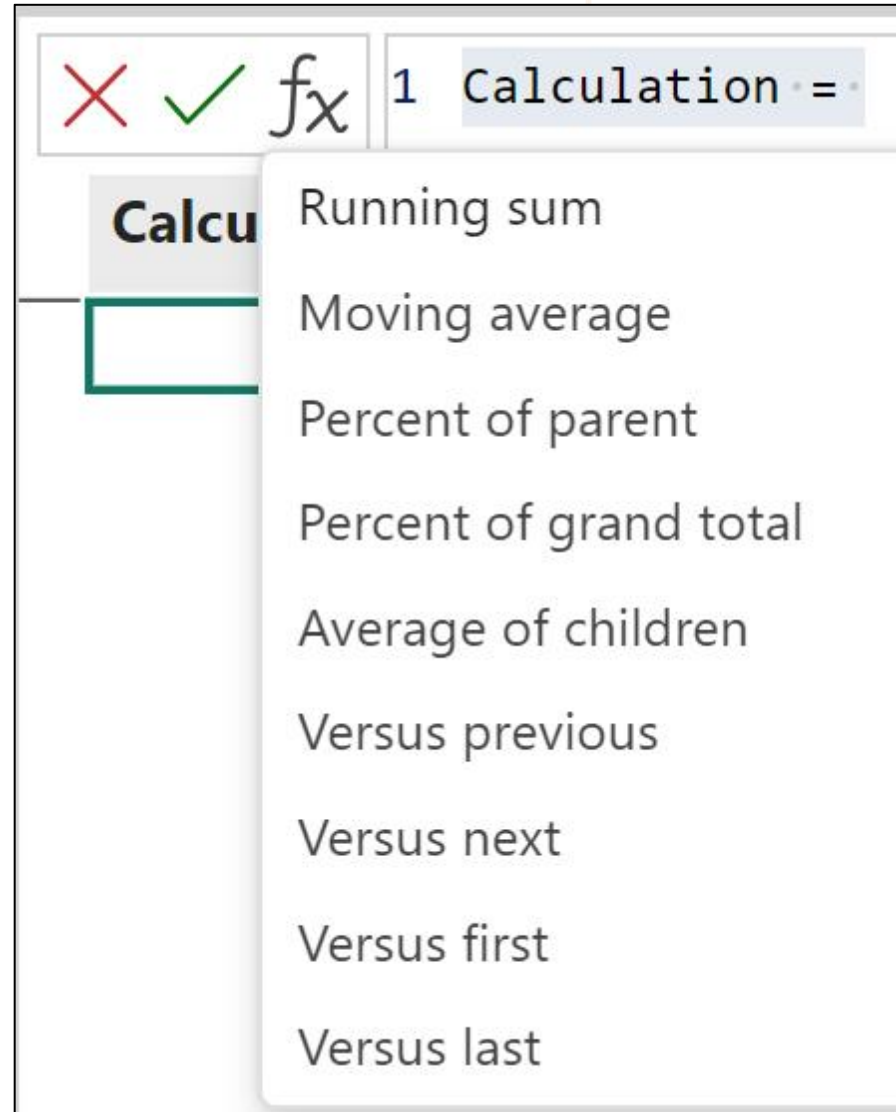
- Start with a visual
- Add new visual calculation
- Add a calculation in formula bar, such as:
$$\text{Profit} = [\text{Sales Amount}] - [\text{Total Product Cost}]$$

✕ ✓ f_x 1 Profit = [Sales Amount]-[Total Product Cost]			
Fiscal Year	Sales Amount	Total Product Cost	Profit
FY2018	\$23,860,891.17	\$20,824,957.60	3,035,933.57
FY2019	\$34,070,108.50	\$30,362,108.34	3,708,000.16
FY2020	\$51,878,274.54	\$46,070,842.02	5,807,432.52
Total	\$109,809,274.20	\$97,257,907.95	12,551,366.25



Use templates

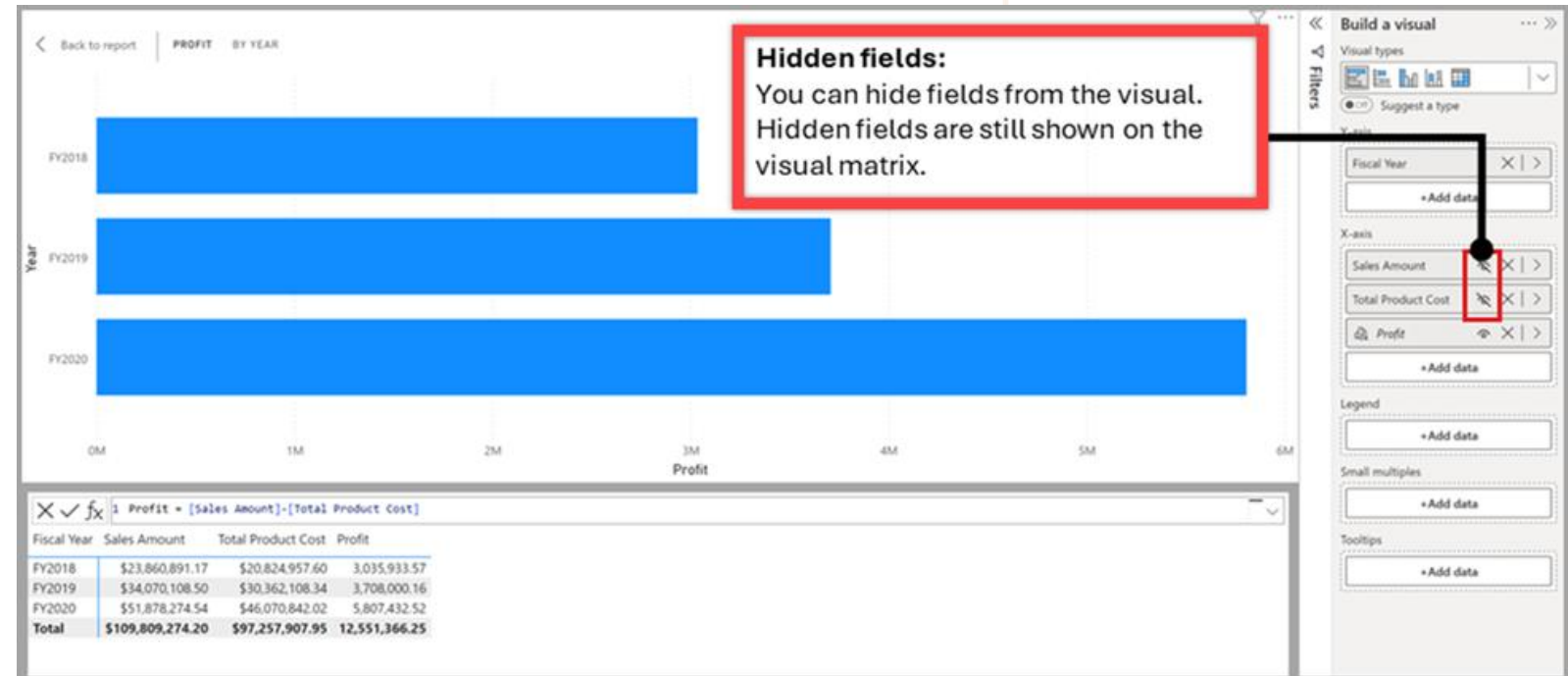
Templates reduce code complexity by using a single function to perform complex calculations.





Hide visual calculations

- Hide fields in the visual matrix when:
 - The visual calculation depends on the field.
 - The field doesn't make sense for your visual.





Use the Axis parameter

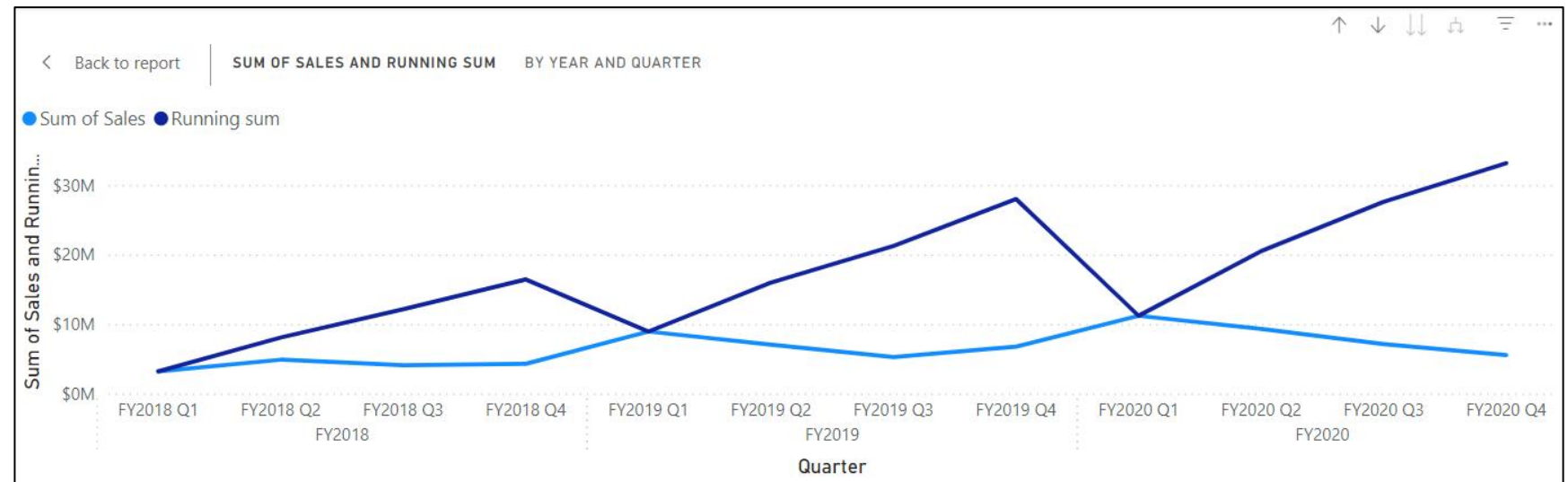
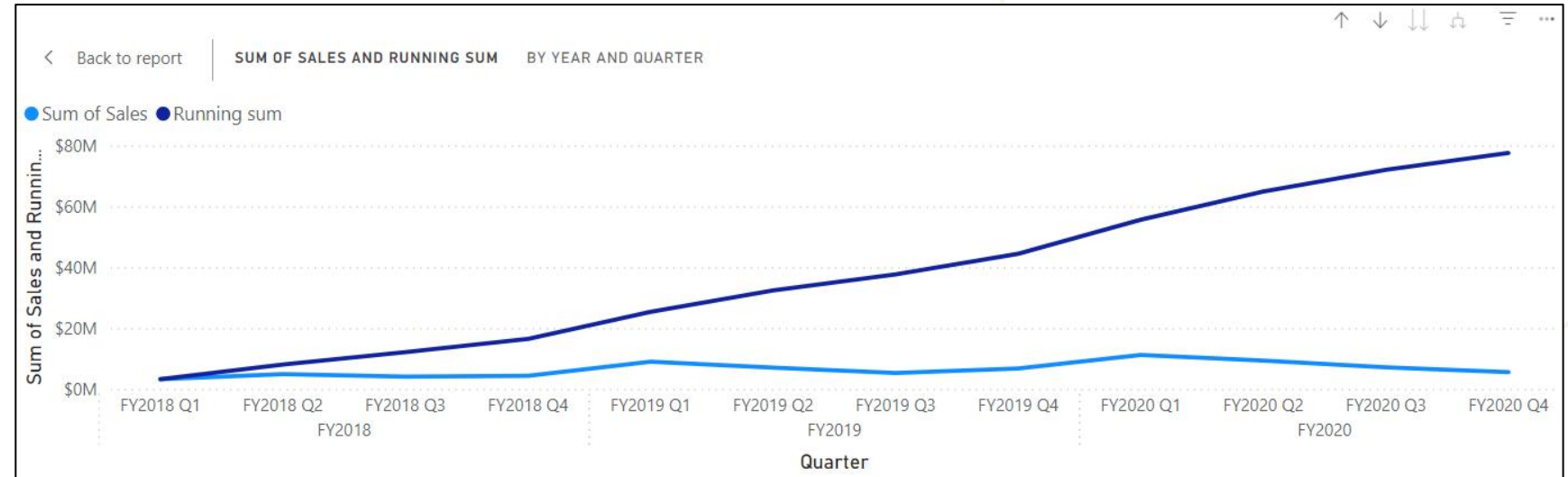
- Determines evaluation direction

Parameters	Icons
ROWS	
COLUMNS	
ROWS COLUMNS	
COLUMNS ROWS	



Use the Reset parameter

- Doesn't reset by default
- HIGHESTPARENT
- LOWESTPARENT

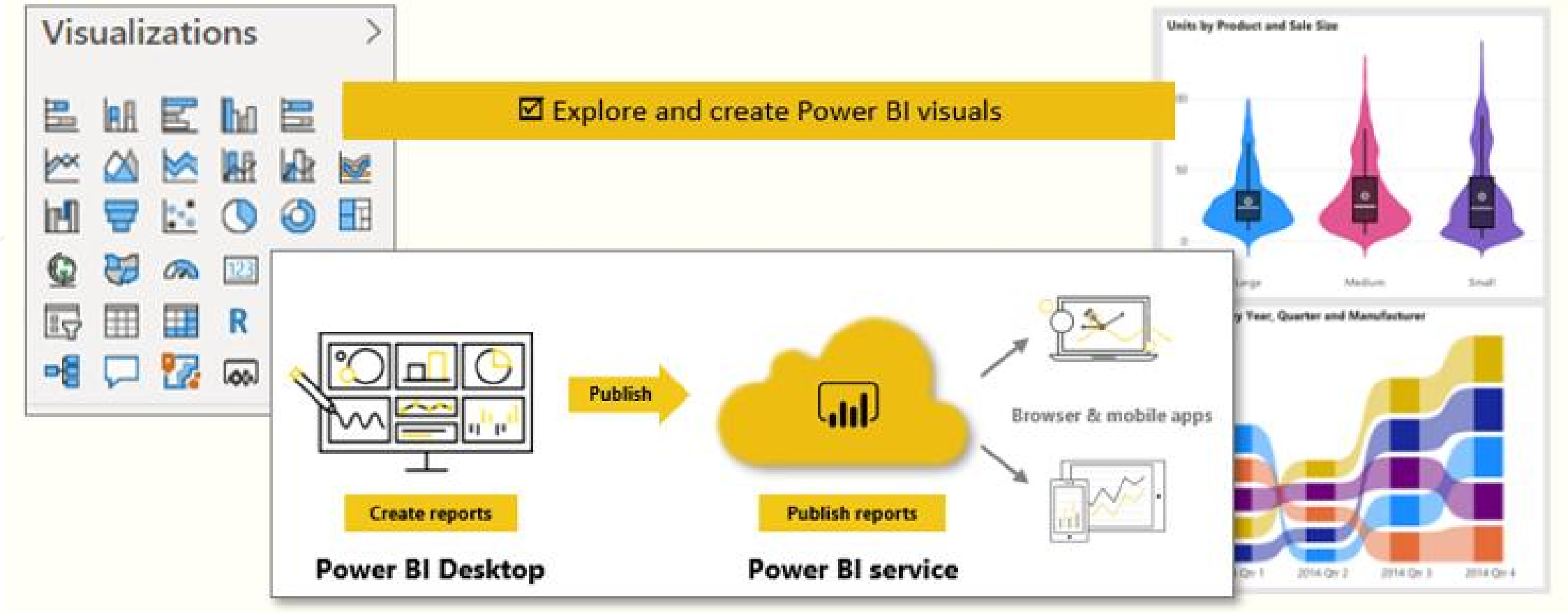


Design Power BI reports





Add visualizations to reports

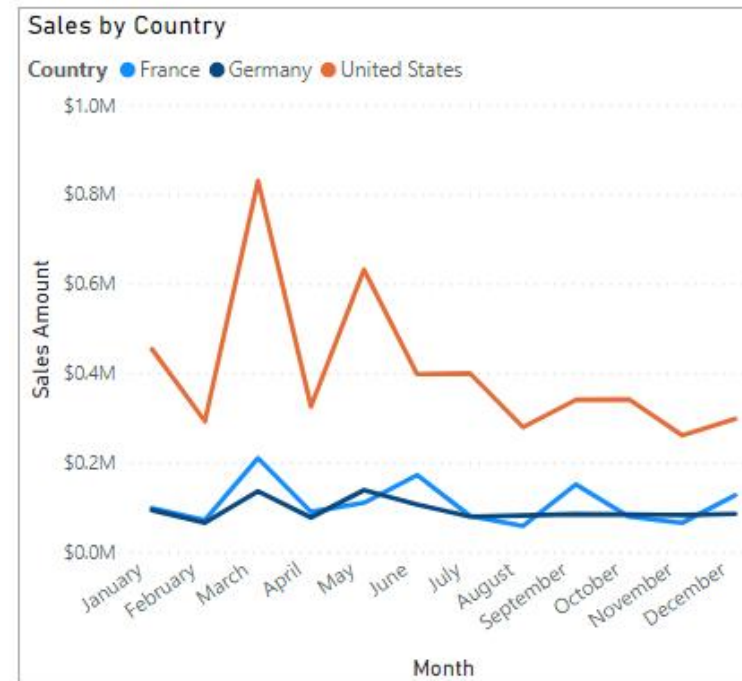




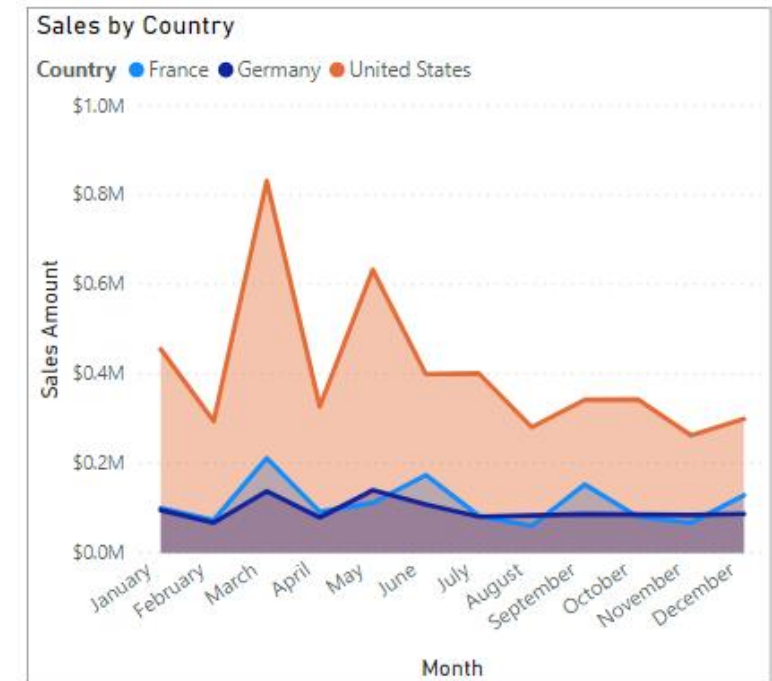
Choose effective visualizations

Country	Sales Amount
United States	\$7,390,464
France	\$773,445
Germany	\$493,628
Total	\$8,550,077

Table



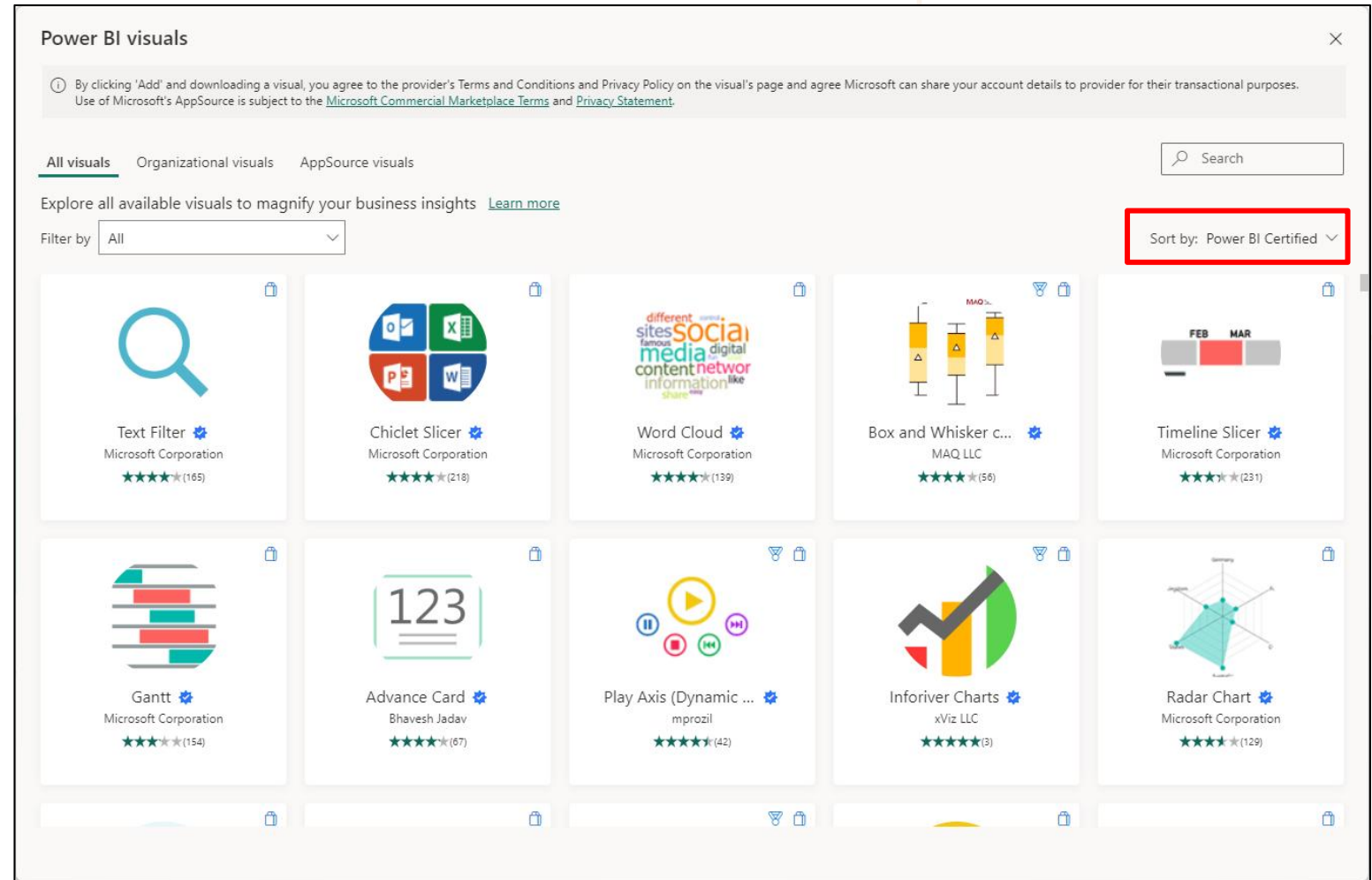
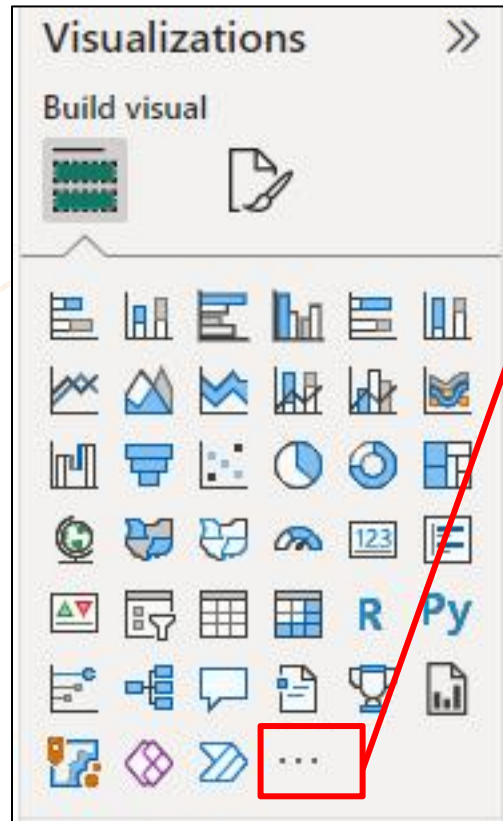
Line Chart



Area Chart



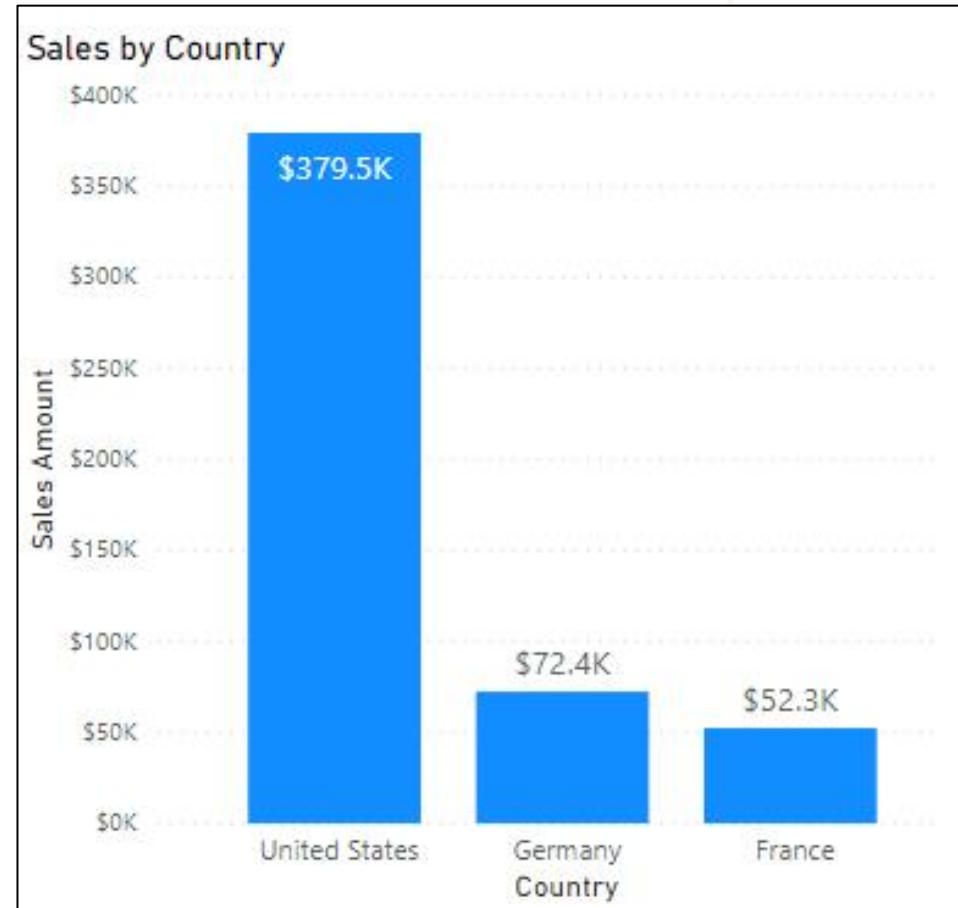
Import custom visuals from AppSource



Format and configure visualizations

Common options:

- Title
- Background
- Border
- Font type/size/color
- Data labels
- Tooltips



Visualizations

Format visual

Search

Visual General ...

> X-axis On

> Y-axis On

> Secondary y-axis Off

> Legend On

> Small multiples

> Gridlines

> Zoom slider Off

> Columns

> Lines

> Markers Off

> Data labels Off

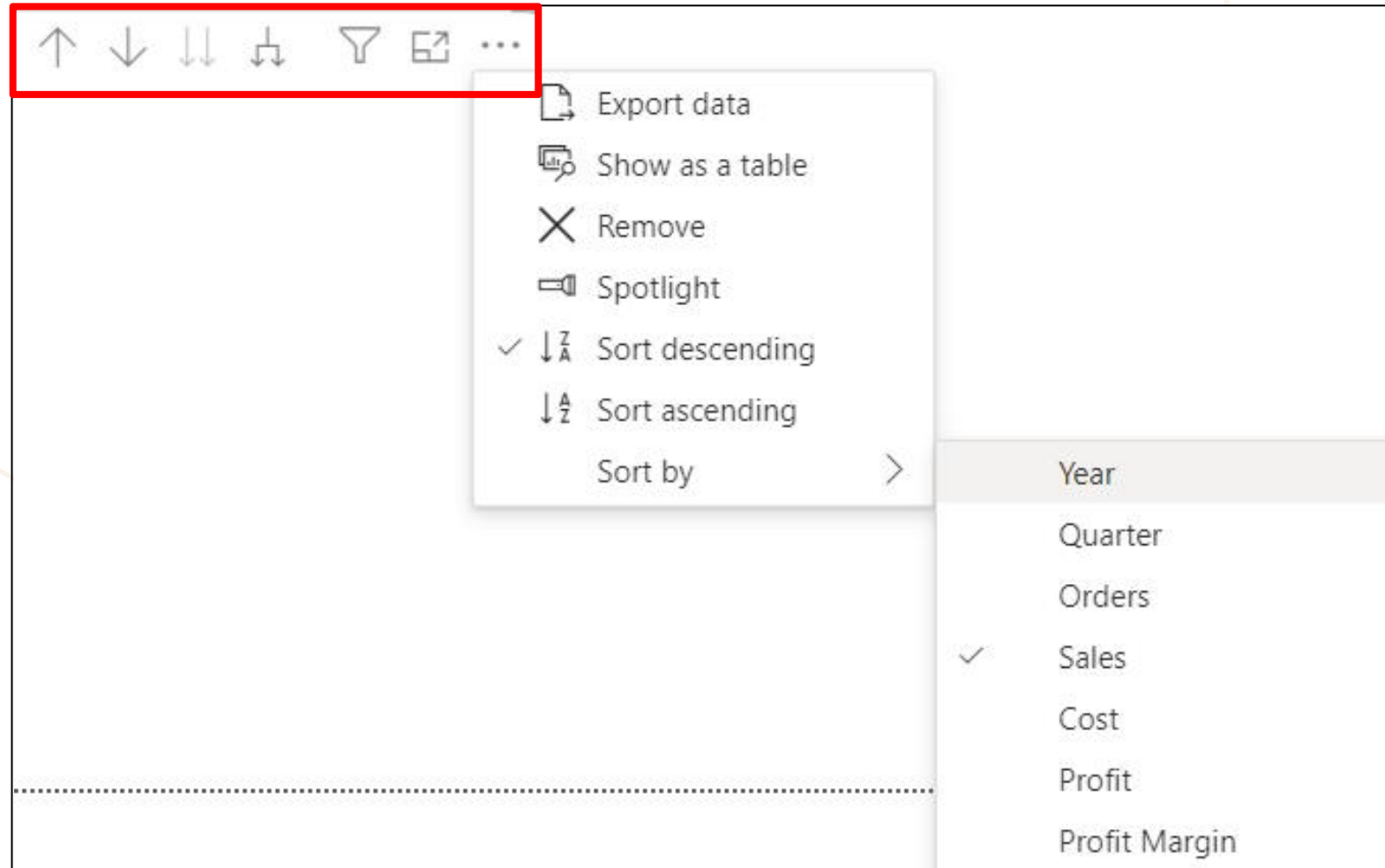
> Series labels Off

> Total labels Off

> Plot area background



Sort visuals in the most effective order





Configure conditional formatting

- Enhance certain visuals with dynamic color schemes.

Visual General ...

Values

> Values

Reset to default

> Column headers

> Totals

> Specific column

Cell elements

Apply settings to

Series

Sales

Background color

On

fx

Background color - Background color

Format style

Gradient

Apply to

Values only

What field should we base this on?

Sum of Sales

Summarization

Sum

How should we format empty values?

As zero

Minimum

Lowest value

Center

Middle value

Maximum

Highest value

Add a middle color

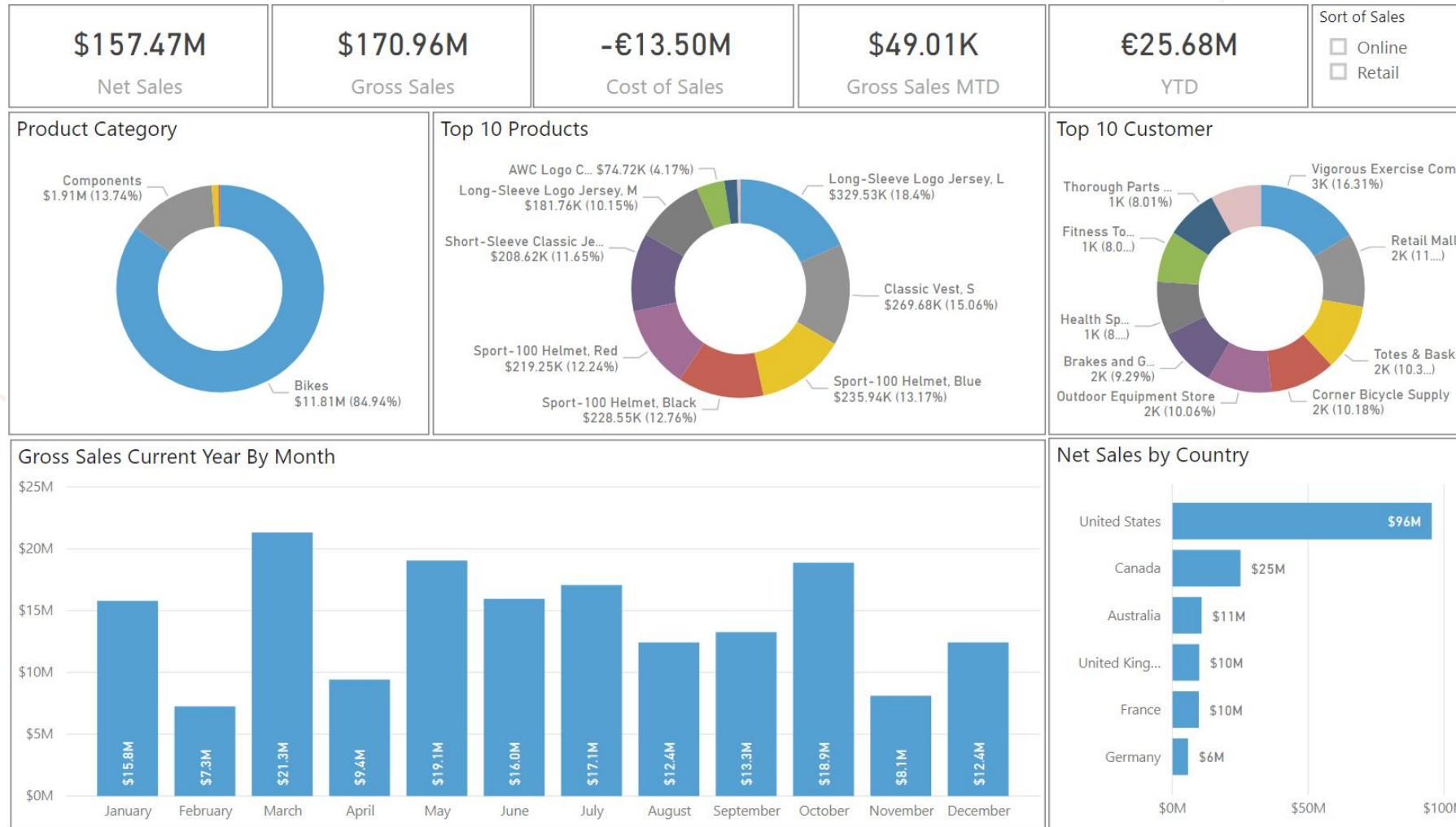
Learn more about conditional formatting

OK Cancel

Region	Sales
Australia	\$1,391,025
Canada	\$13,875,633
Central	\$7,633,387
France	\$4,527,840
Germany	\$1,877,743
Northeast	\$6,715,354
Northwest	\$12,004,822
Southeast	\$7,638,607
Southwest	\$18,001,116
United Kingdom	\$3,883,043
Total	\$77,548,570

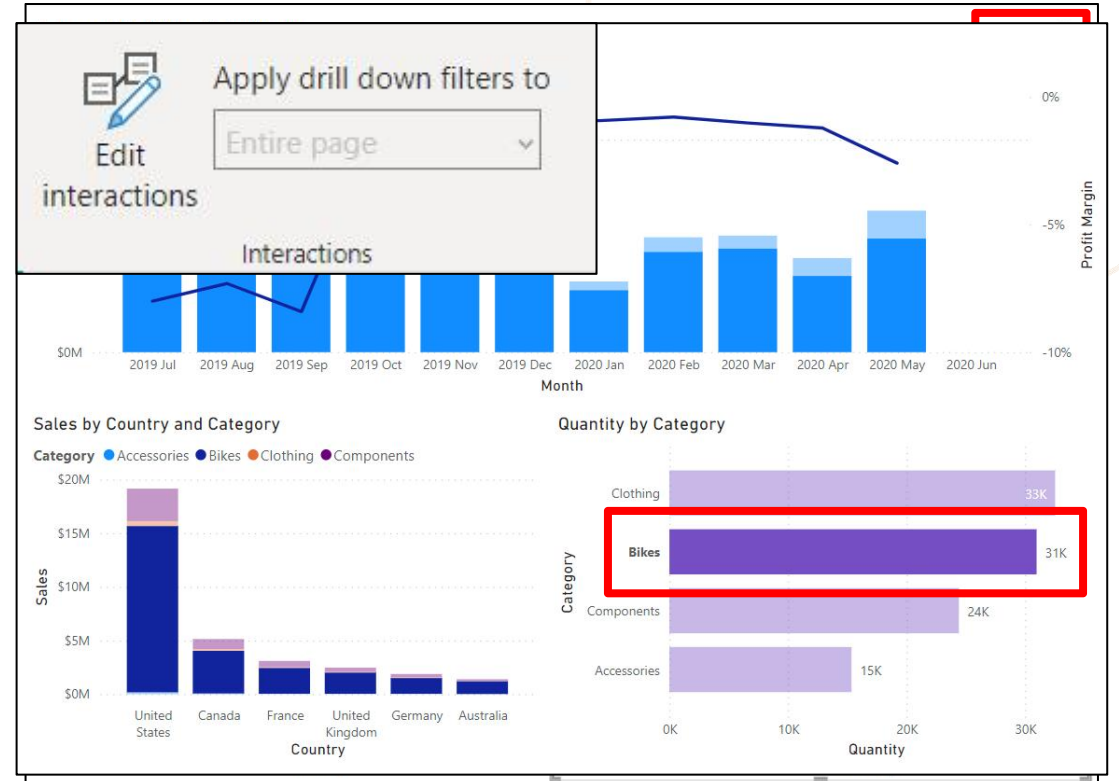
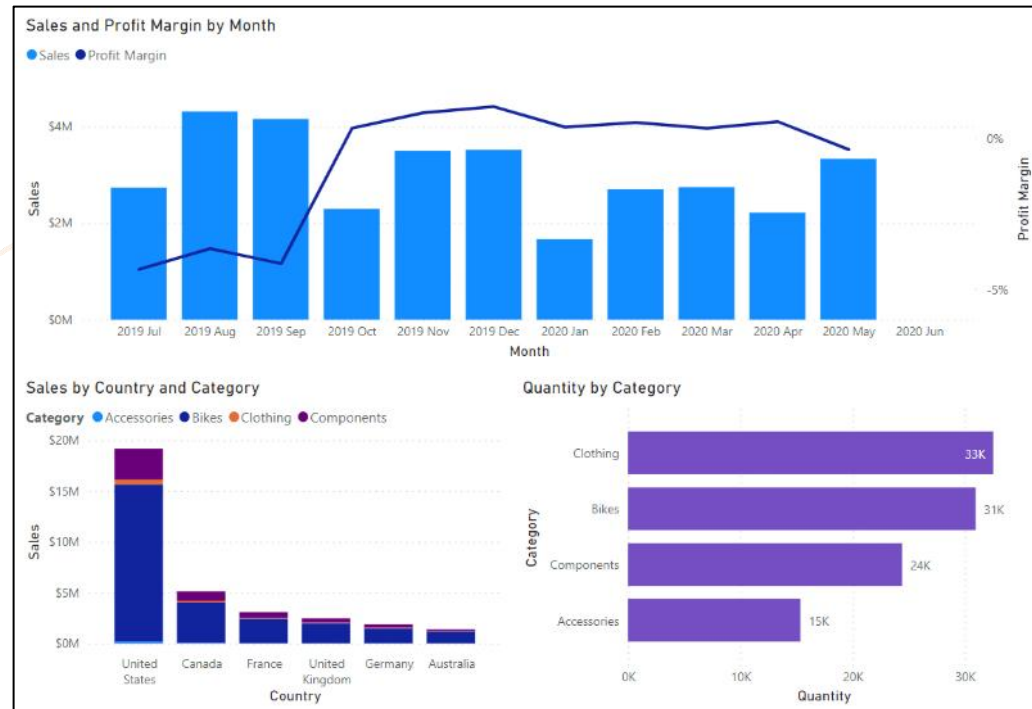


Design an appealing report layout



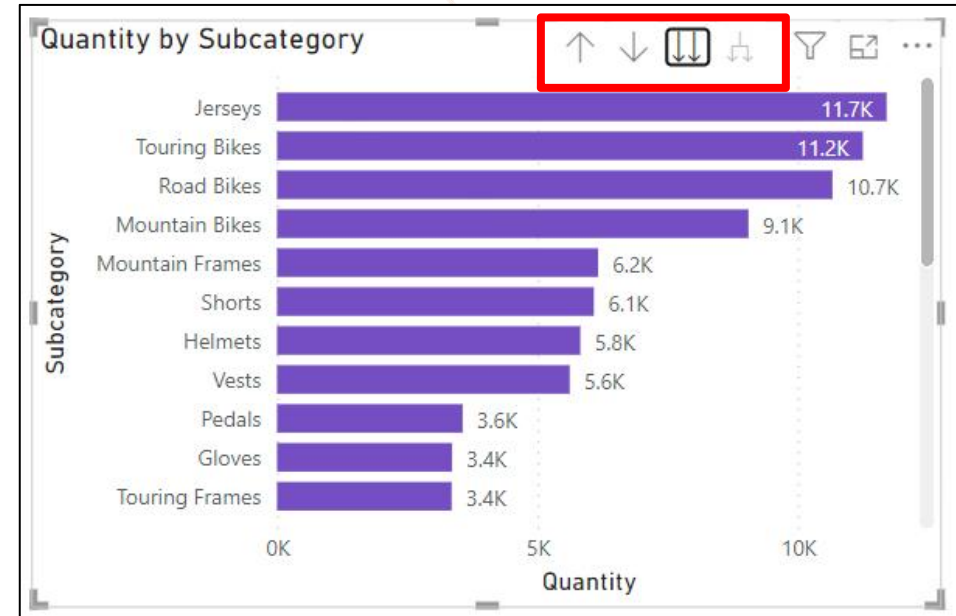
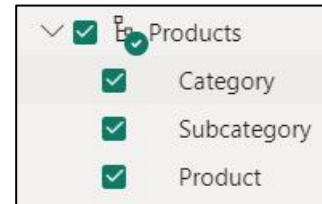
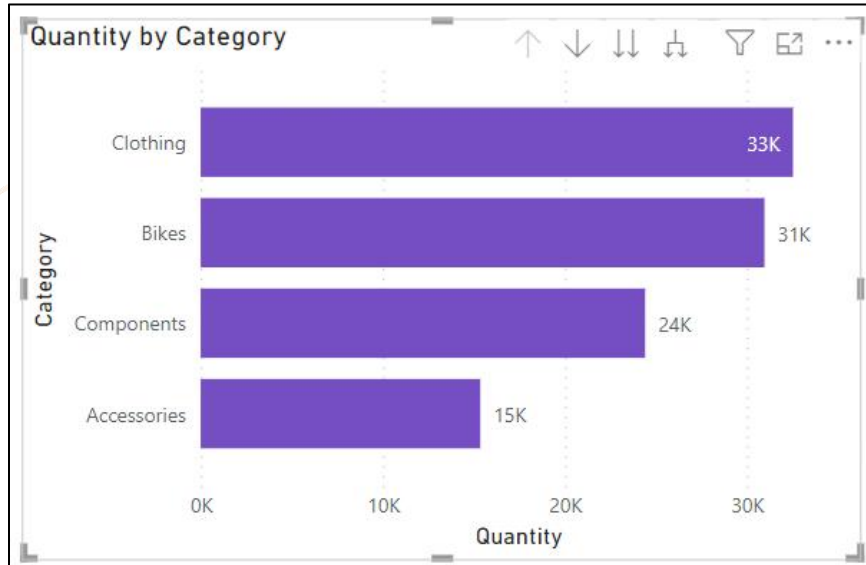


Cross-highlight between visuals





Drill down through a hierarchy in a visual

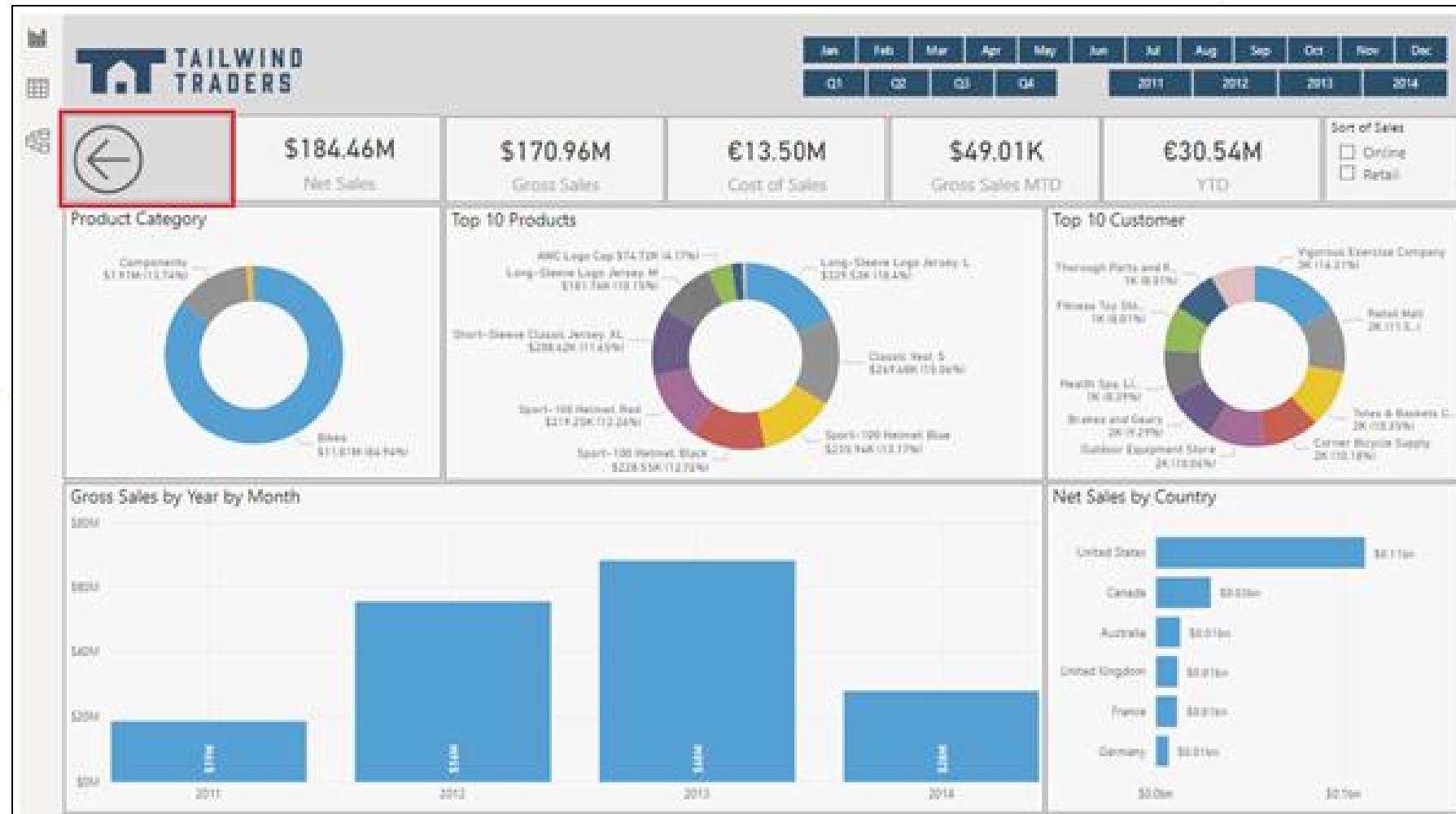


Enhance reports for user
experience



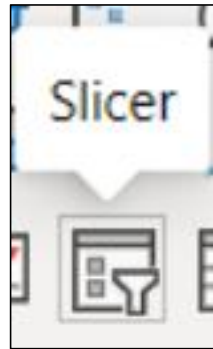


Design functional report navigation





Use the Slicer visual to provide on-report filtering



Year ▼

FY2020 ▼

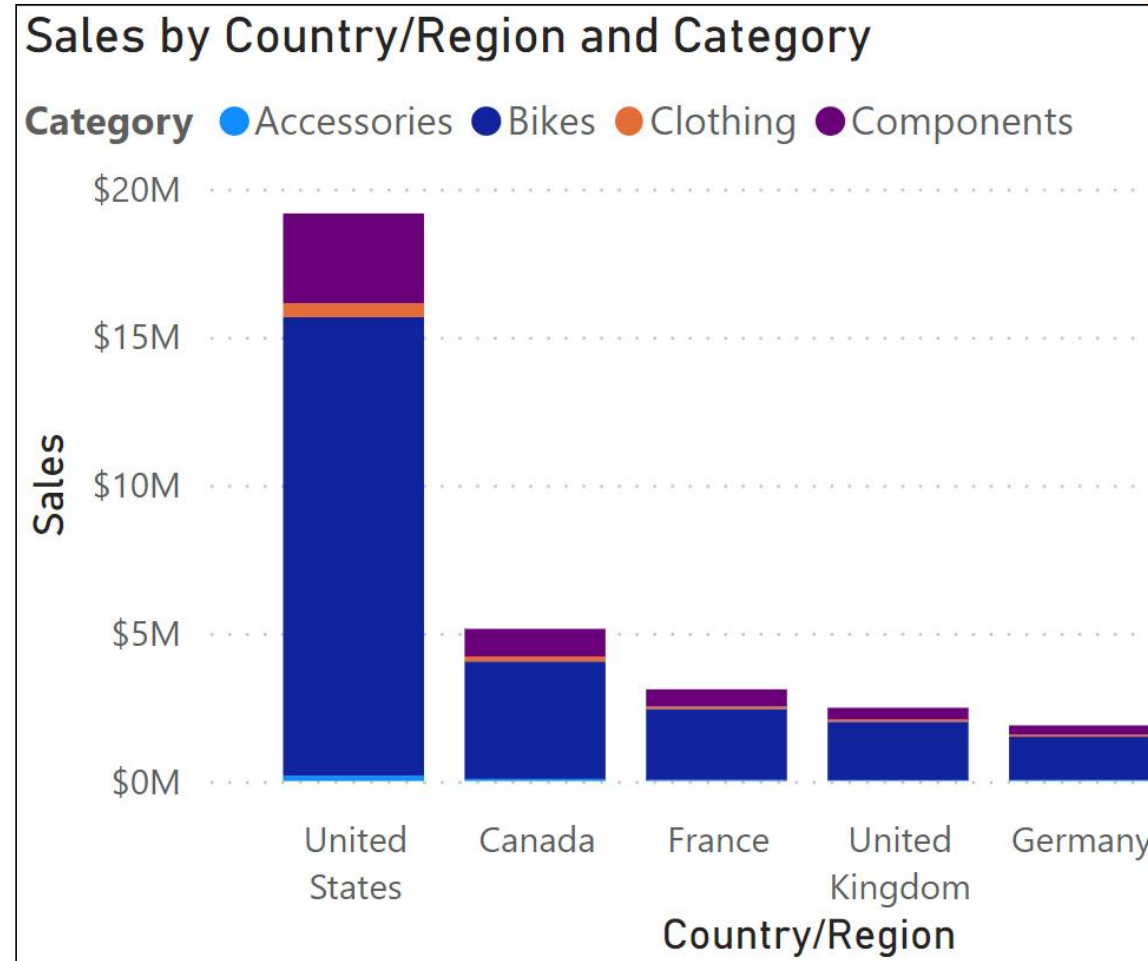
Group, Country, Region

☐ Select all

▼ ☐ Europe

▼ ☒ North America

▼ ☐ Pacific



Sync slicers » ×

Add and sync with all pages, or select specific pages:

Page name	↺	👁
Overview	☒	☒
Profit	☒	☒
My Performance	☐	☐
Product Details	☐	☐

▼ Advanced options

Enter a group name to sync selection to any other visuals with that group name

Region

☒ Sync field changes to other slicers

☒ Sync filter changes to other slicers



Create complex filter conditions with the Filter pane

Filters

Search

Filters on this visual

Add data fields here

Filters on this page

Add data fields here

Filters on all pages

Add data fields here

Date

Date is (All)

Filter type ⓘ

Basic filtering

Advanced filtering

Basic filtering

Relative date

Relative time

Top N

Profit Margin is (All)

Show items when the value

is less than

☒ And ☐ Or

Apply filter

Filters

Search

Subcategory is (All)

Year is (All)

Filter type ⓘ

Basic filtering

Search

☐ Select all

☐ FY2018 365

☐ FY2019 365

☐ FY2020 366

☐ FY2021 365

☐ FY2022 365

☐ Require single selection

Visualizations

Data

Search

> Targets

> Date

☐ Date

> ☒ Fiscal

☐ Month

☐ Quarter

☐ Year

> Product

> Region

> Reseller

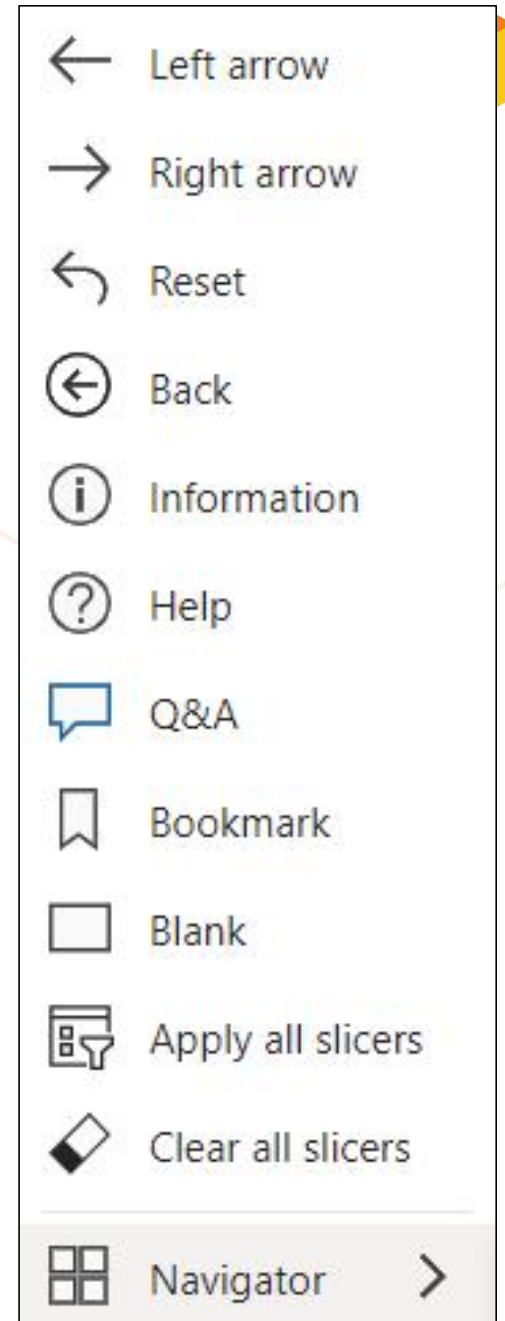
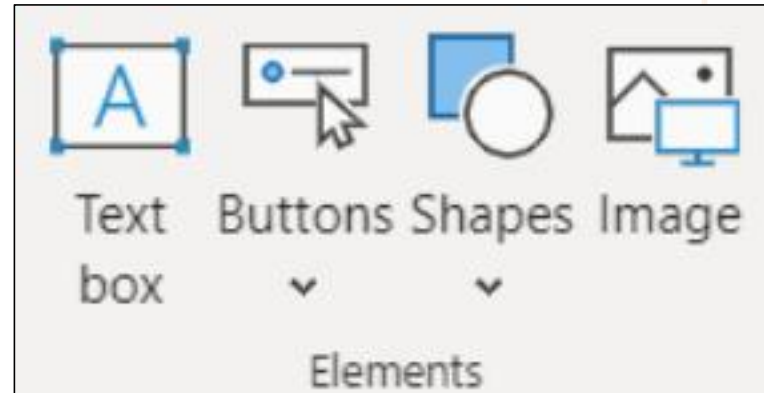
> Sales

> Salesperson

> Salesperson (Performan...

Add shapes and buttons

- Enhance reports with:
 - Text boxes
 - Images
 - Shapes
-
- Improve navigation with Buttons.





Preserve report state with bookmarks

- Bookmarks capture:
 - Current page
 - Filters
 - Slicers
 - Sort order
 - Drill location

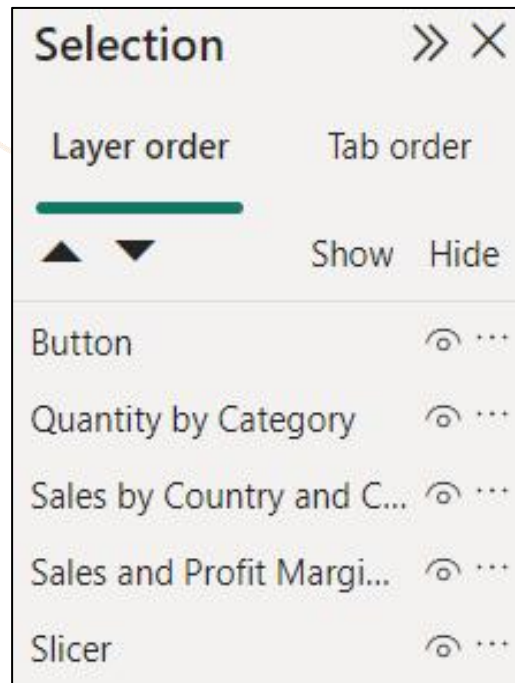




Use selection order for accessibility

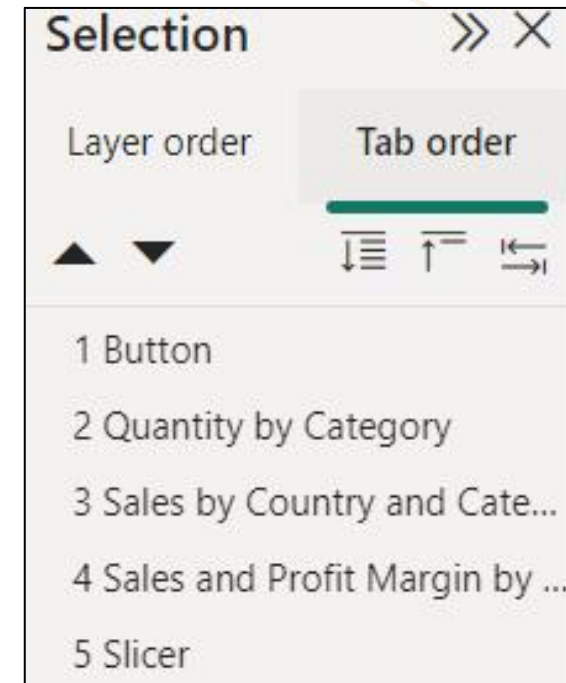
Layer order:

Defines the order of visuals.



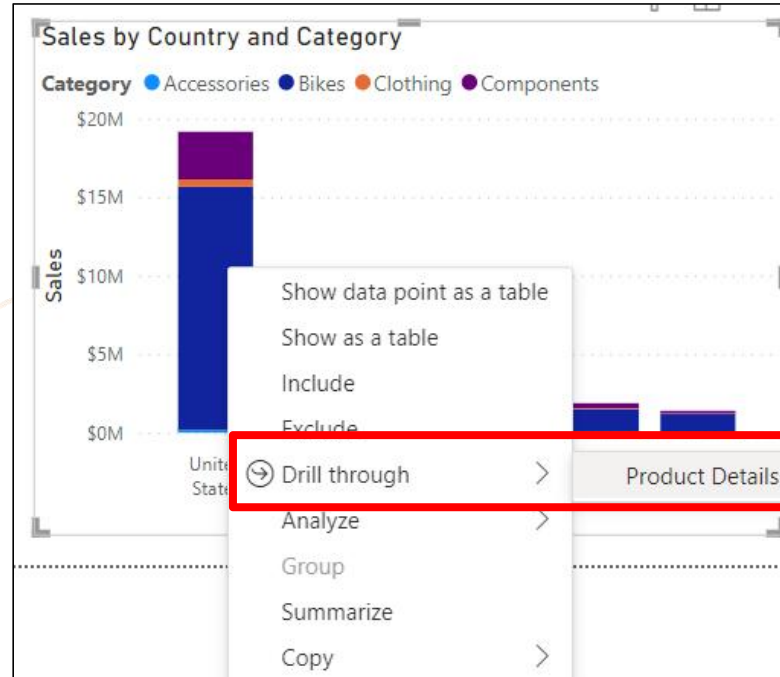
Tab order:

Controls keyboard user navigation.





Drill through from one visual to another



Bikes				
Subcategory	Color	Quantity	Sales	Profit Margin
Mountain Bikes	Black	2,501	\$2,484,496	8.69% ●
Mountain Bikes	Silver	3,384	\$2,603,065	7.56% ●
Road Bikes	Black	3,040	\$2,389,419	-6.13% ◆
Road Bikes	Red	205	\$300,532	-6.07% ◆
Road Bikes	Yellow	3,938	\$3,380,758	-6.60% ◆
Touring Bikes	Blue	2,988	\$2,627,332	-5.65% ◆
Touring Bikes	Yellow	2,065	\$1,696,351	-16.63% ◆
Total		18,121	\$15,481,954	-2.62%

Drill through

Cross-report ☐ Off

Keep all filters ☒ On

Category is Bikes

Country is United States

Year is FY2020



Choose when to use paginated reports

Buying Group

Tailspin Toys

Location

Absecon, NJ,Aceituna...

[View Report](#)

Invoices From Date:

05/01/2016

Invoices To Date:

05/31/2016

Buying Group Account Statement

Invoice Range: May 01, 2016 to May 31, 2016

Buying Group: Tailspin Toys, 201 Customers Selected

Floor 20, Import Plaza
105 Silk Road
Tradesville
WA 99999



Invoice No: 73507-70431

Invoice 2 of 3

Invoice Date: May 31, 2016

Delivery Time:

Purchase Order No: 16826

Received By:

Billed To: Tailspin Toys (Head Office)

Delivery Information: Suite 150, 1959 Sarma Road

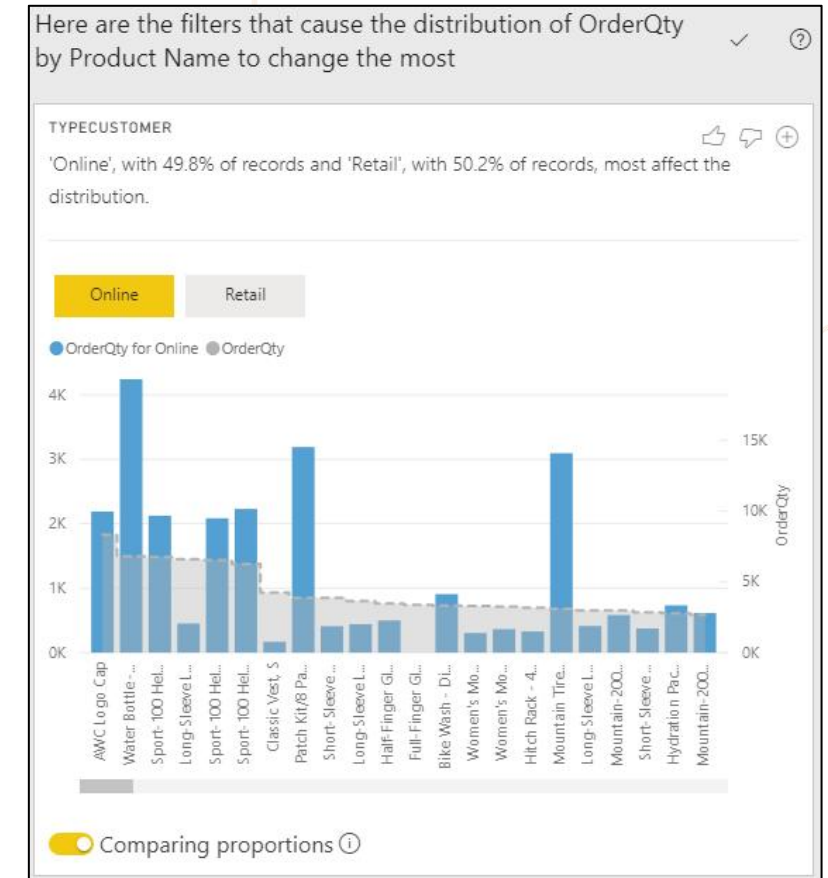
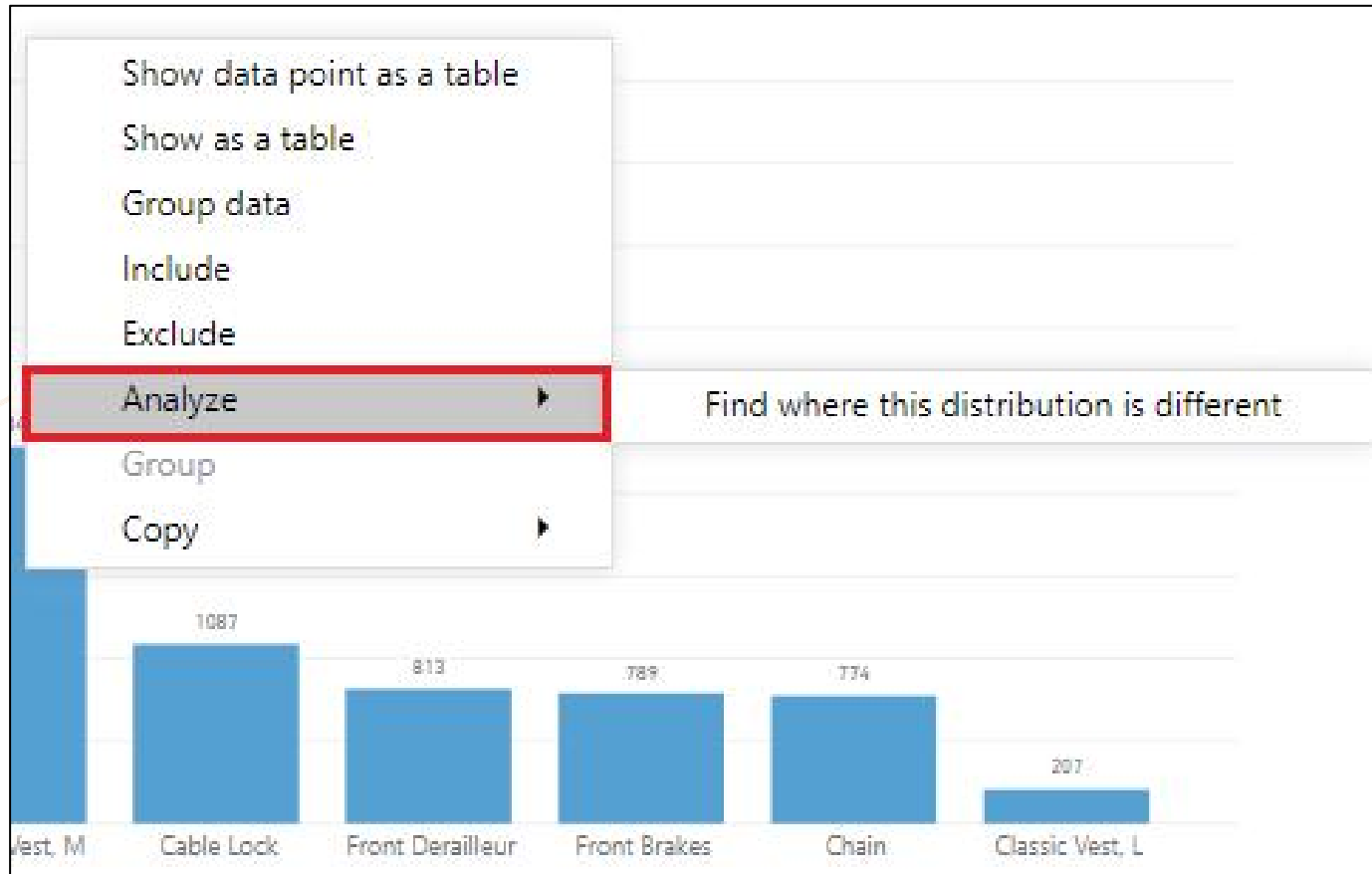
Item	Quantity	Unit Price	Tax	LineTotal
DBA joke mug - you might be a DBA if (White)	7	\$13.00	\$13.65	\$104.65

Perform advanced analytics





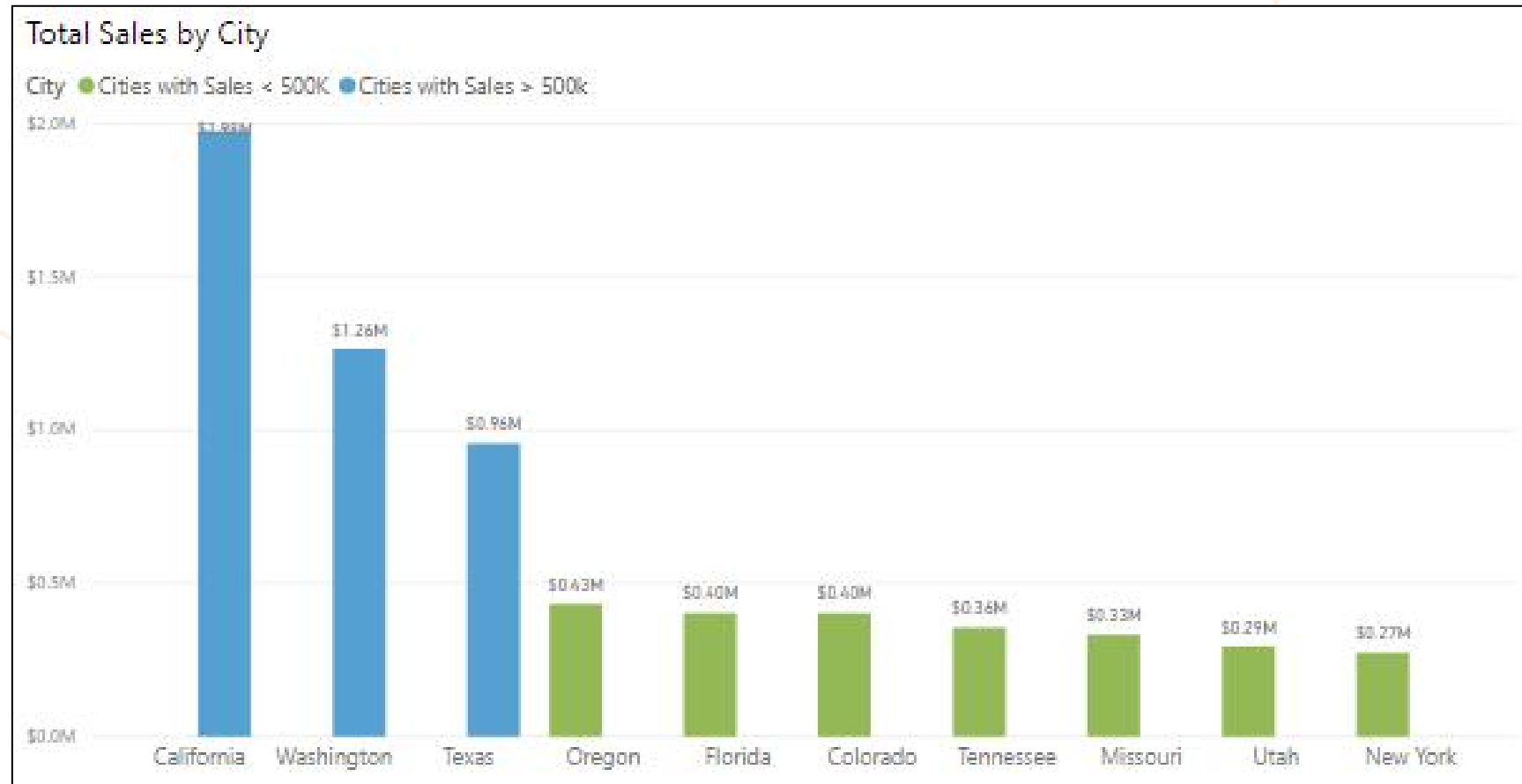
Compare data with the Analyze feature





Group and bin data

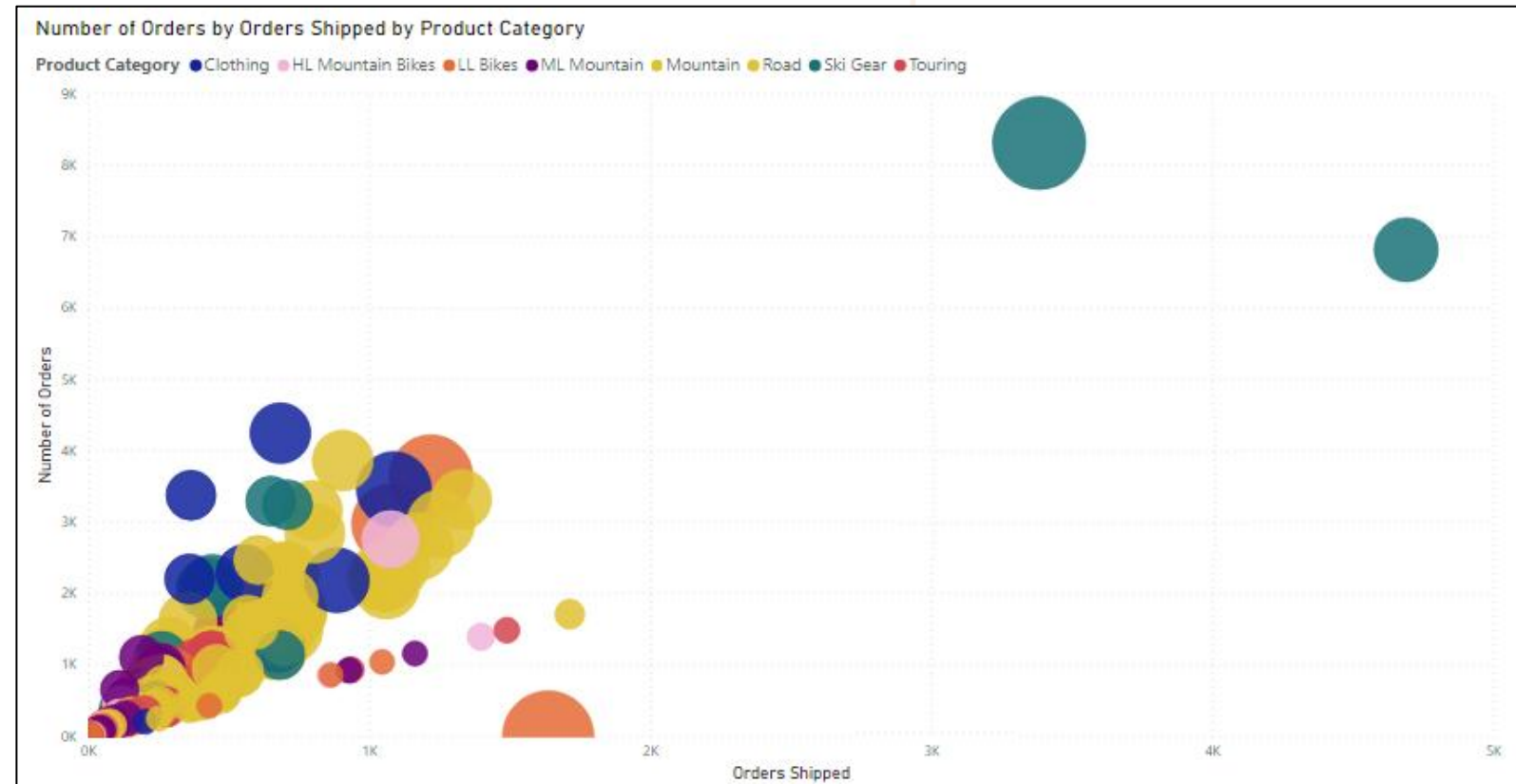
- Organize data into categories or intervals to better analyze and visual data.





Identify outliers in data

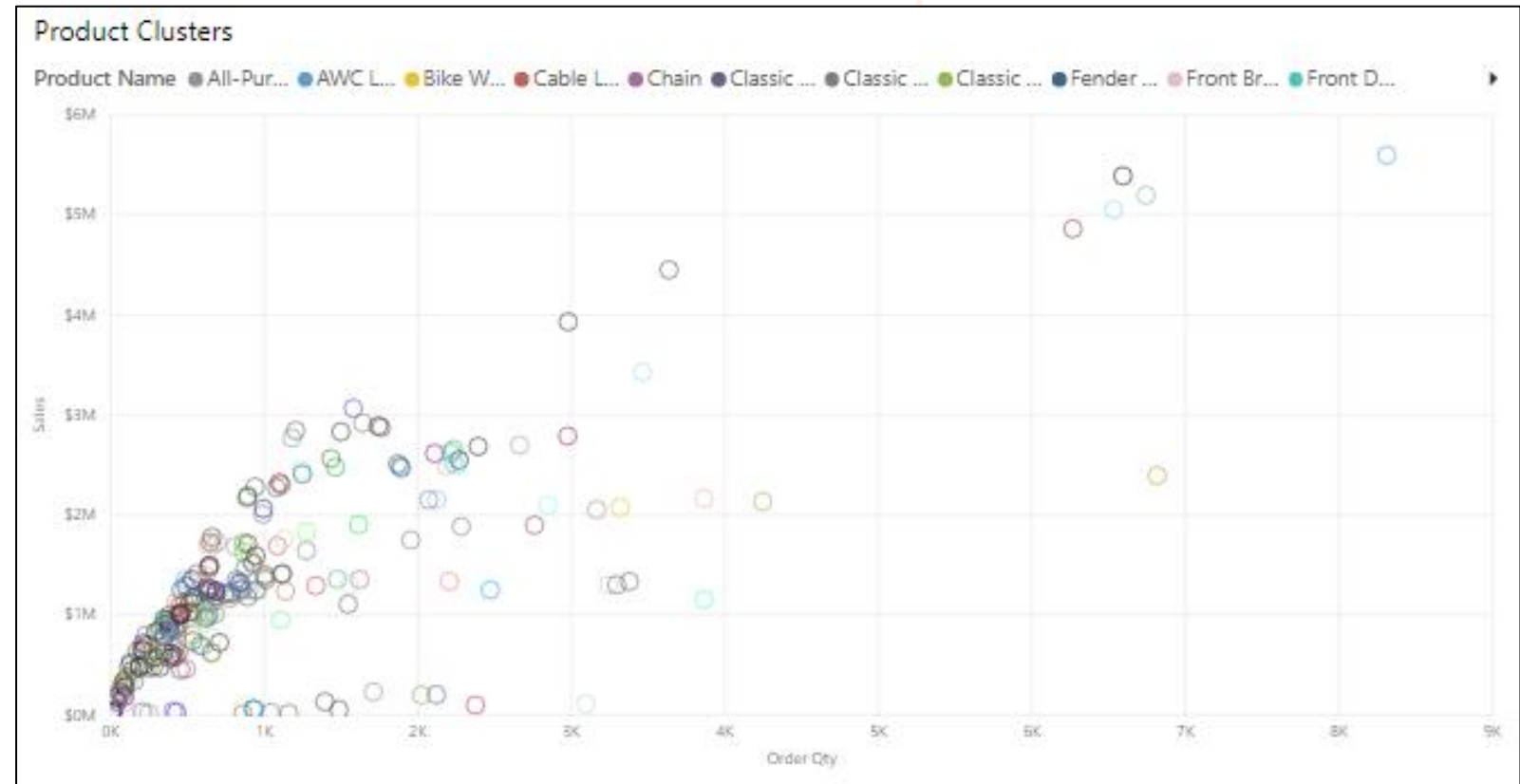
- Unexpected deviations in data.
- Scatter charts can detect outliers.
- Use DAX to define and analyze outliers dynamically.





Clustering with Scatter chart visual

- Automatic data grouping.
- Enhance pattern recognition and insights.
- Simplify data understanding and analysis.





Key Performance Indicator visual

- At-a-glance insights into performance against predefined targets.



Value

Add data fields here

Trend axis

Add data fields here

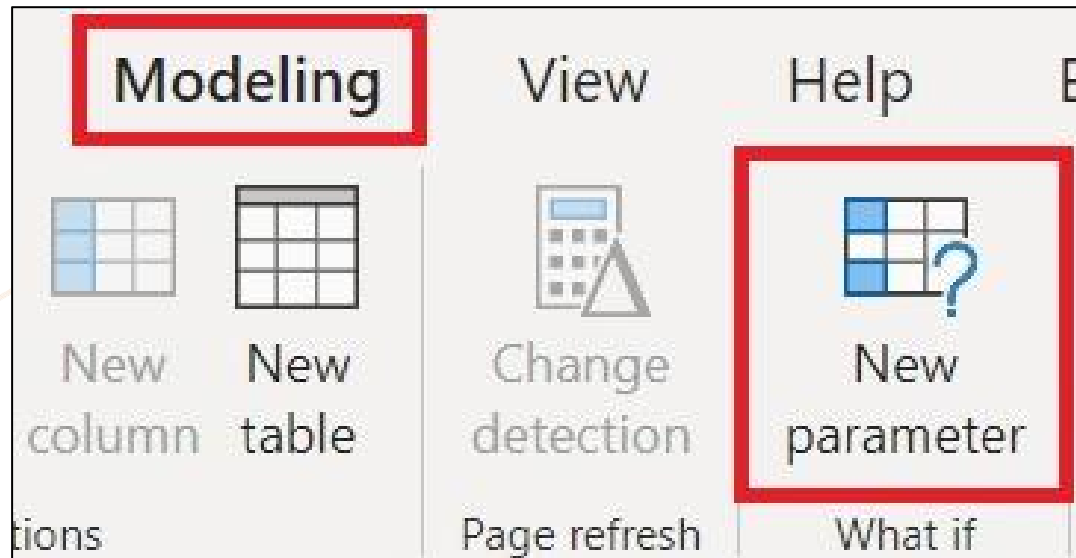
Target

Add data fields here



What-if parameters

- Enable scenario planning and interactive exploration for better decision-making.



What-if parameter

Name

Data type

Minimum

Maximum

Increment

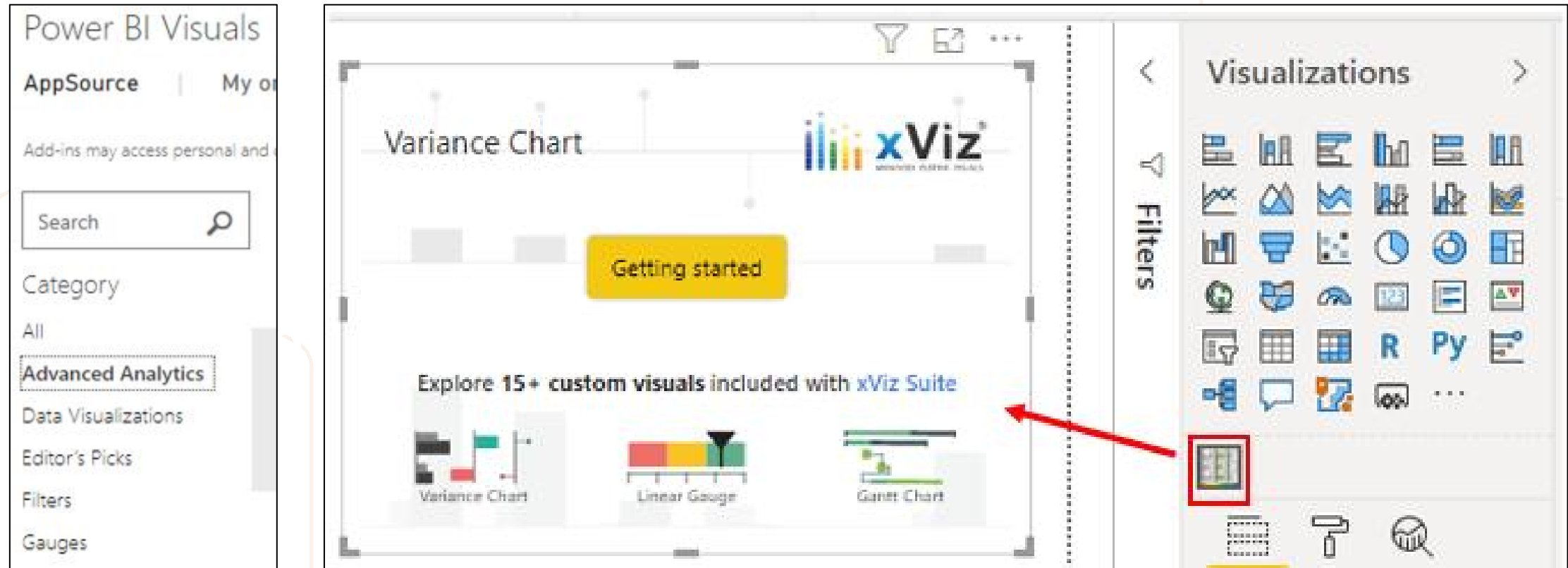
Default

☒ Add slicer to this page



Advanced Analytics custom visuals

- Discover additional visuals developed with Advanced Analytics in mind.





AI Insights for data preparation in Power Query Editor

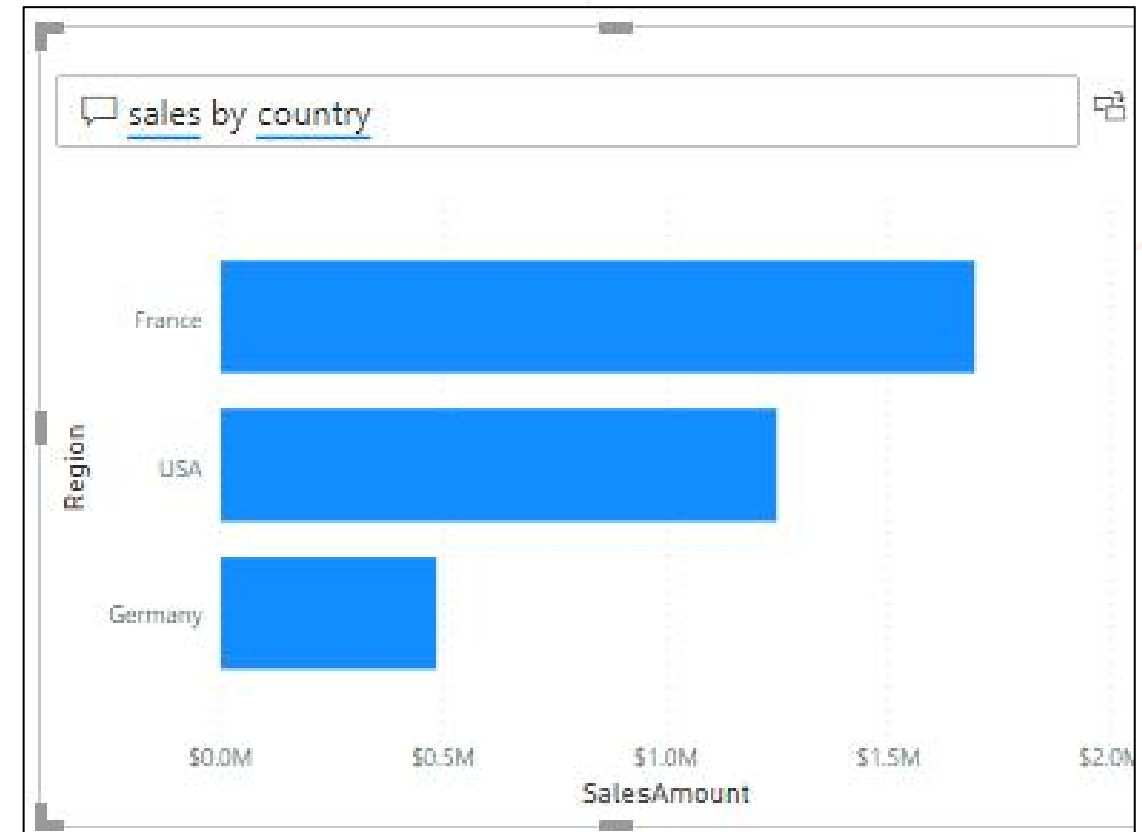
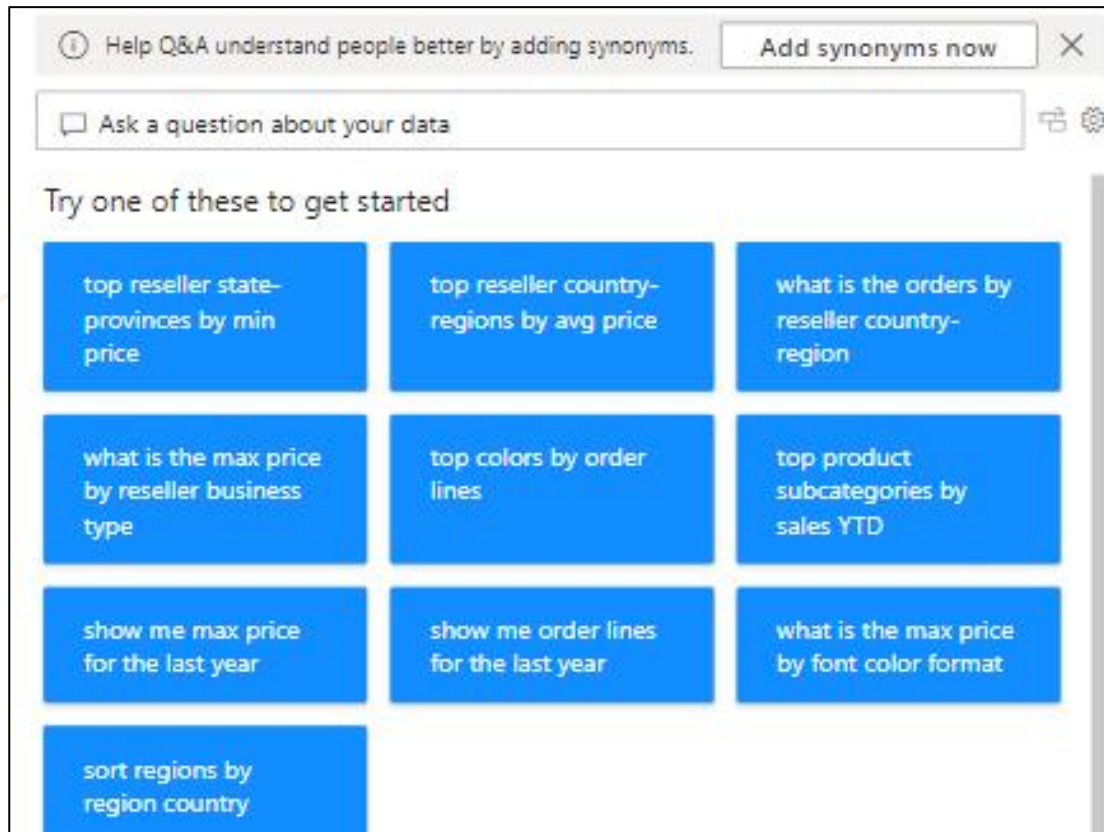
- Leverage AI analysis of text, images, and use Azure Machine Learning models.





AI in the Q&A feature

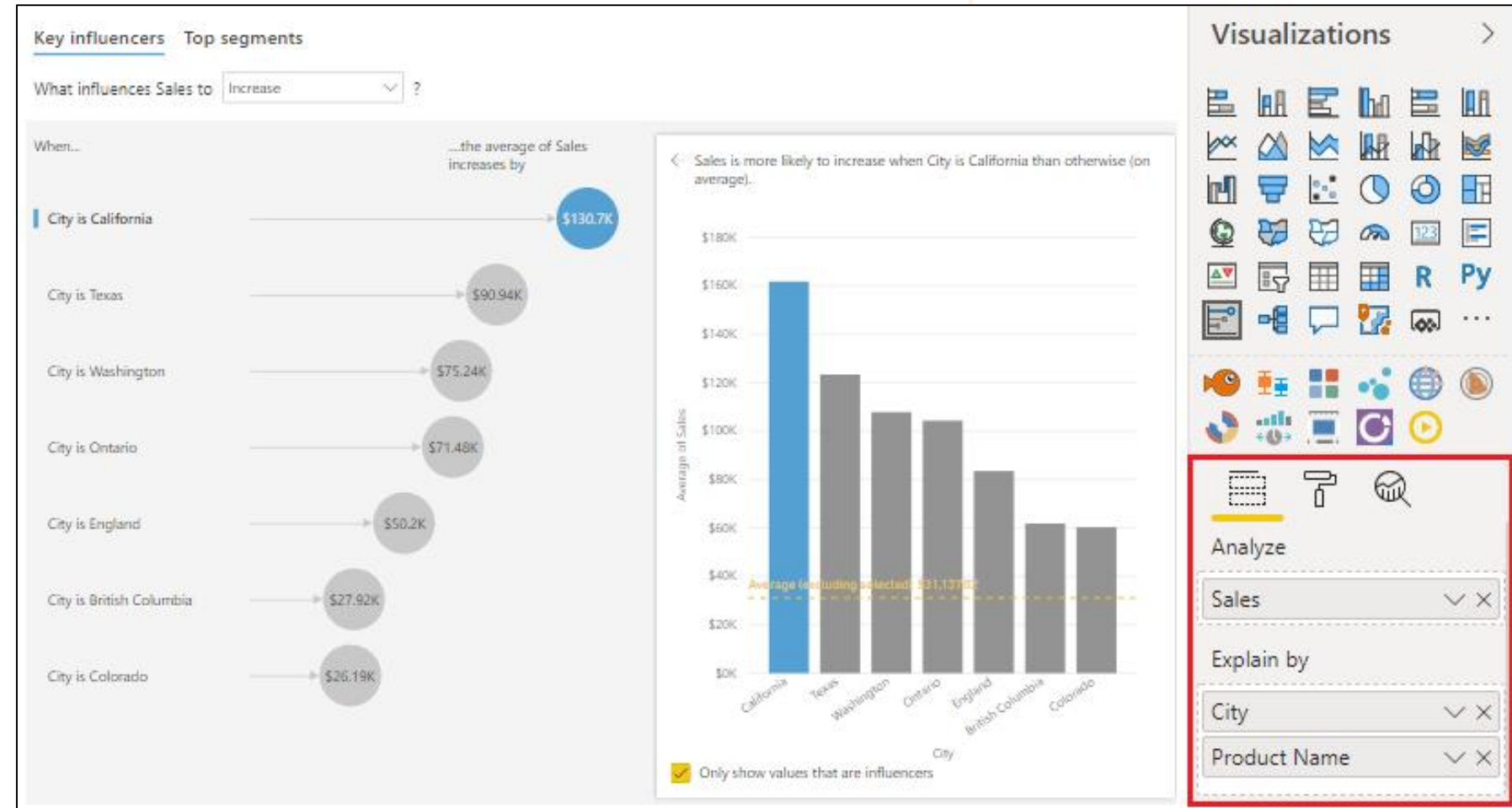
- Use natural language to ask questions about data and create visuals for insights.





Key influencers visual

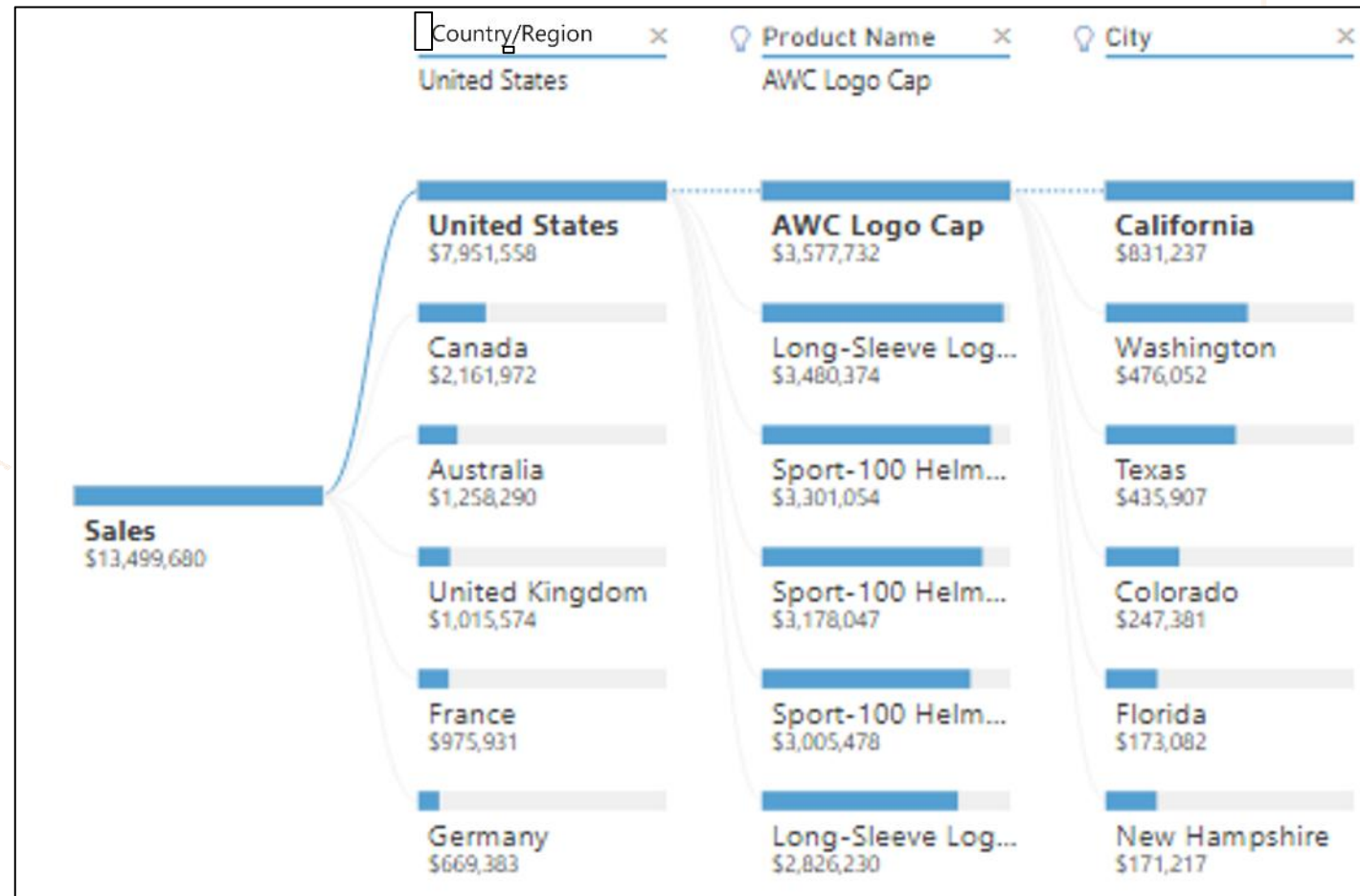
- Identify factors driving variations in a chosen target metric.





Decomposition tree visual

- Visualize data hierarchies for in-depth exploration and pattern discovery.

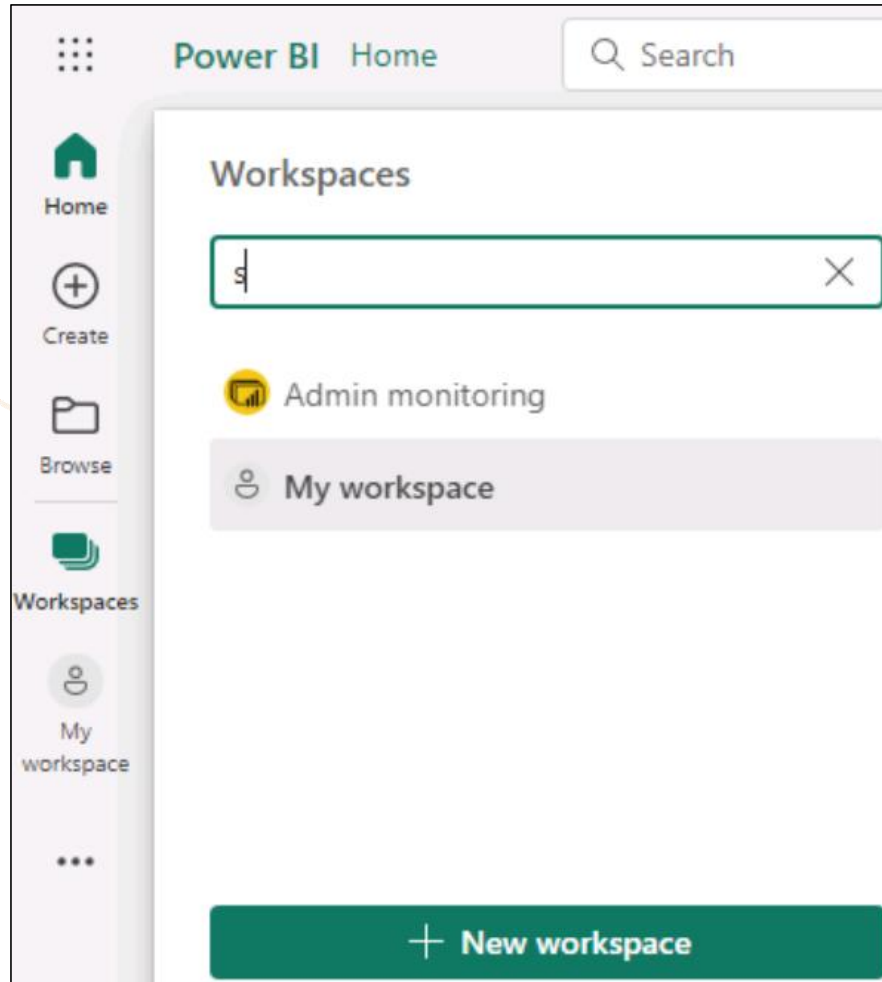


Create and manage workspaces





Create Power BI workspaces



Create a workspace

Name *

Available

Description

Domain (preview) ⓘ

[Learn more about workspace settings](#)

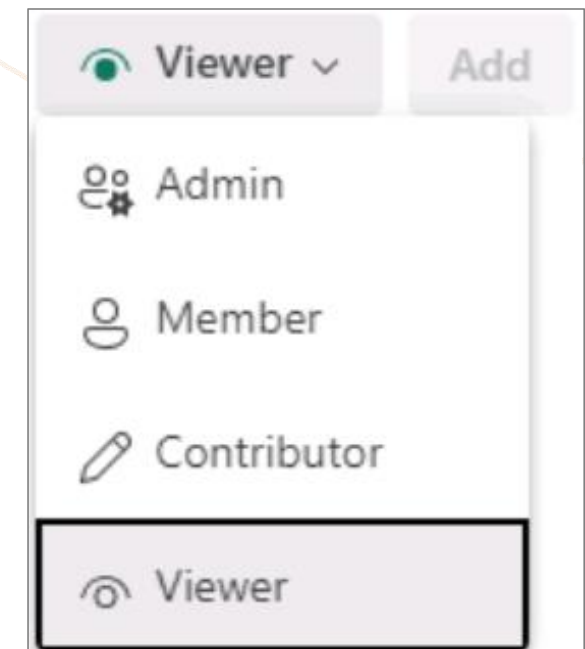
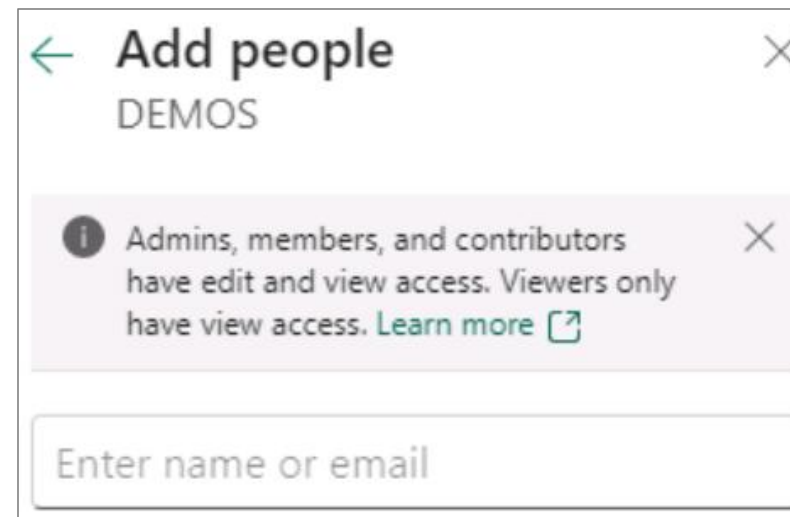
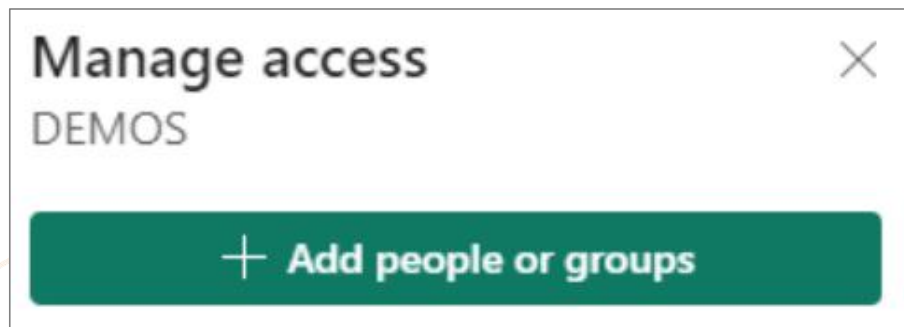
Workspace image

Advanced ▼



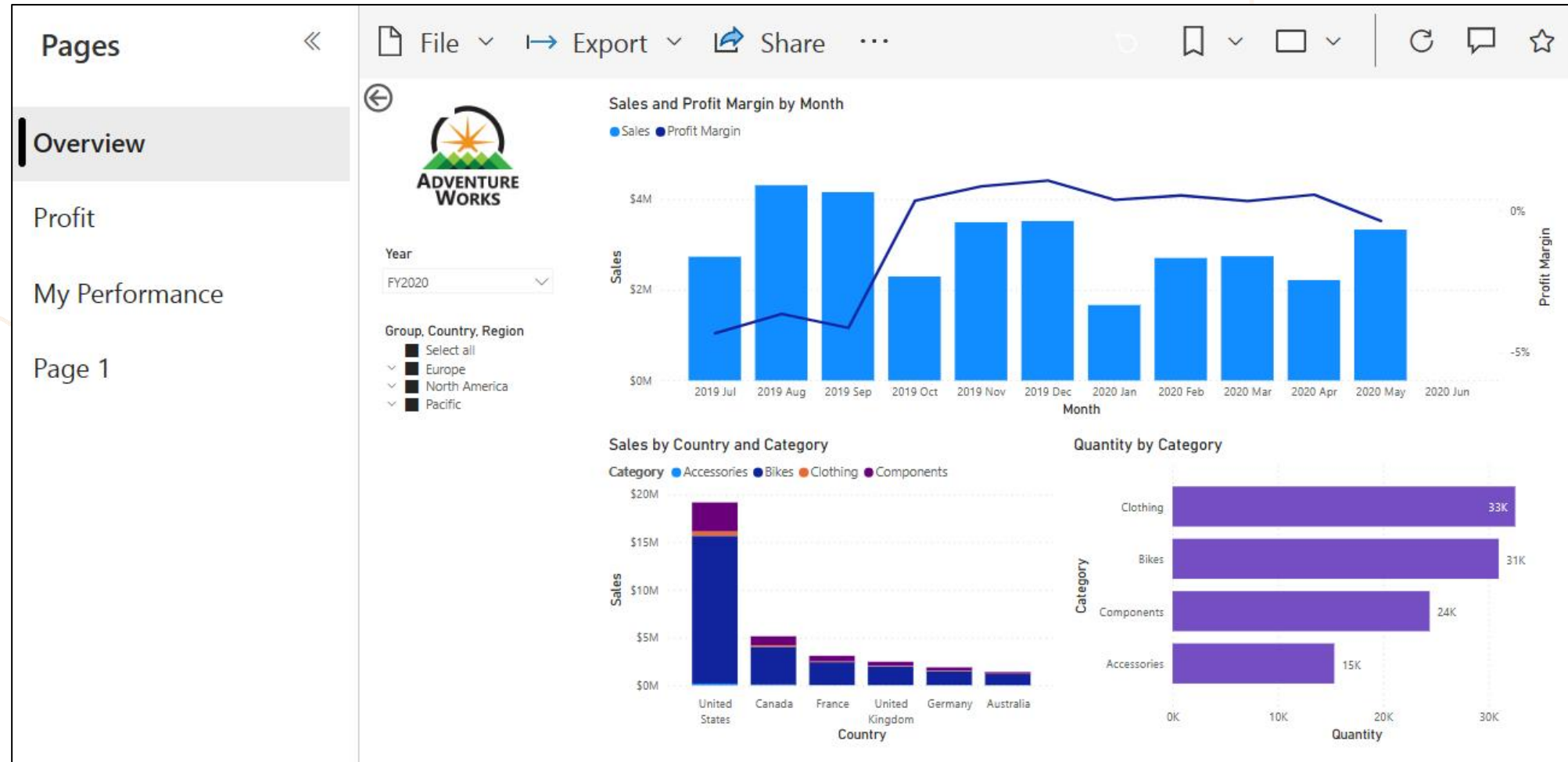
Assign workspace roles

- Use workspaces for collaboration with other developers, not content sharing.



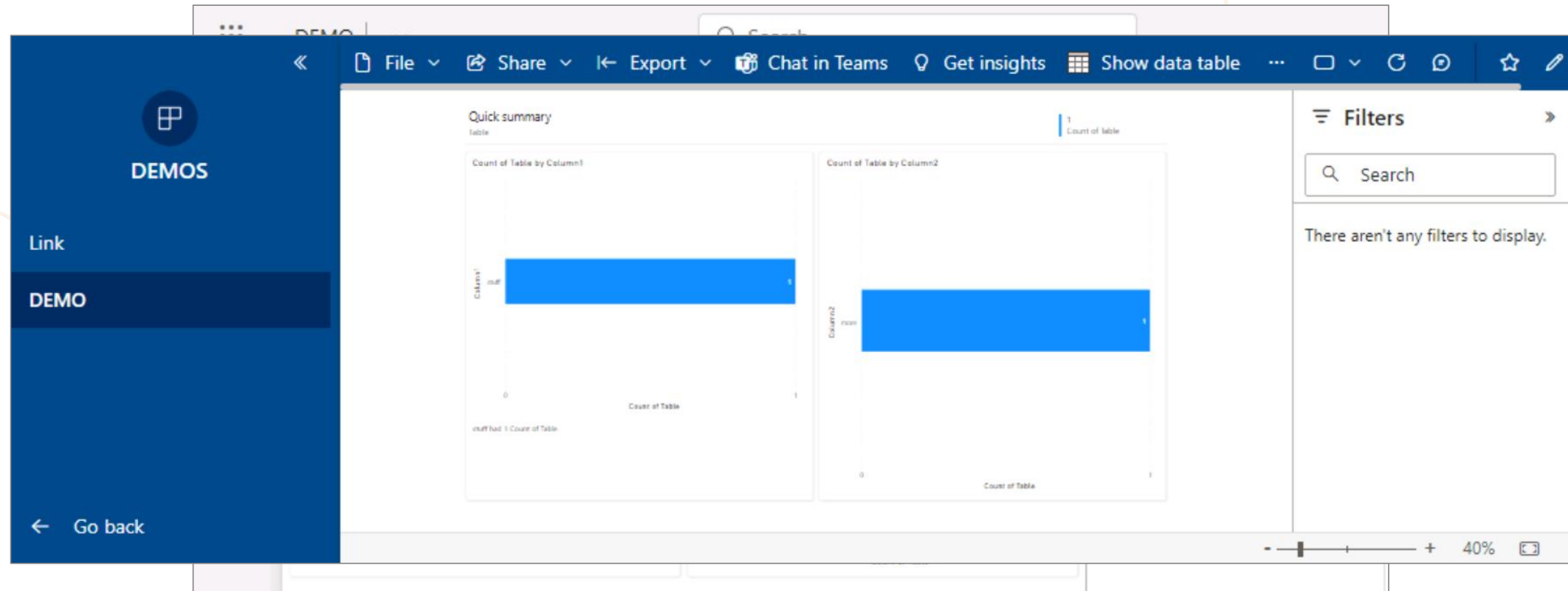


Publish reports to a workspace



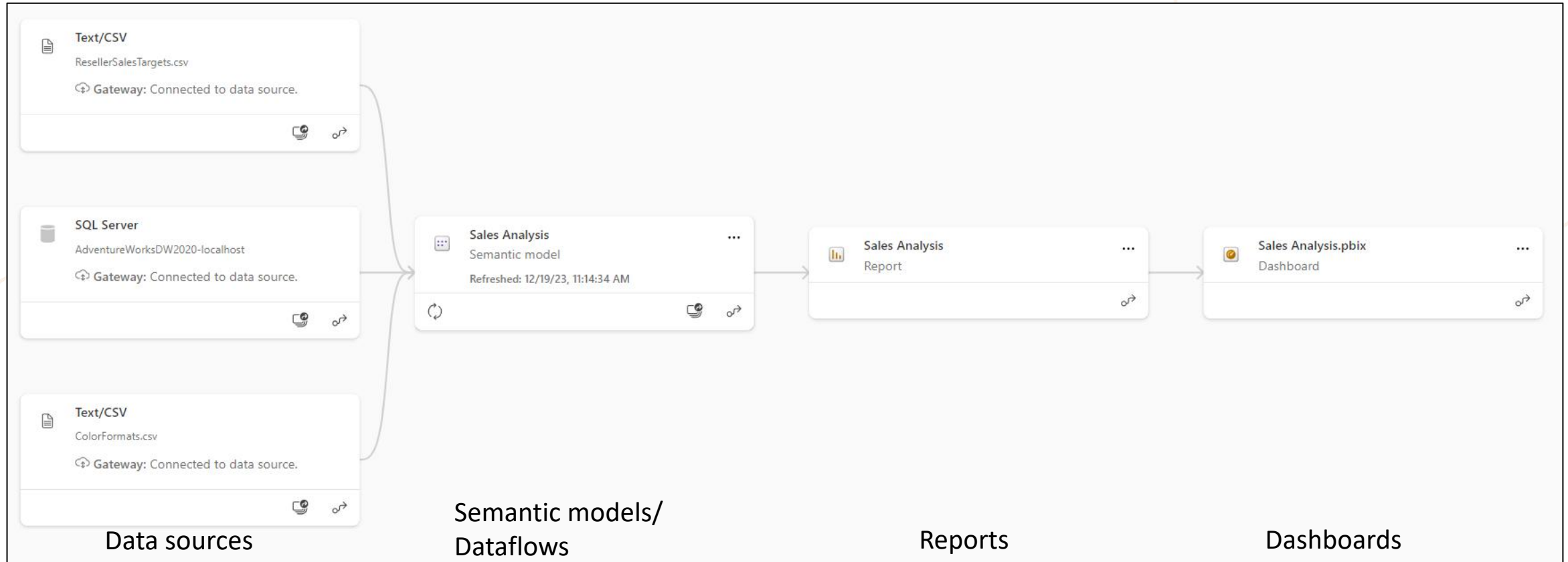


Explore different workspace items





Identify relationships between items with Lineage view





Configure apps for distribution

① Setup ② Content ③ Audience

Create app

Audience

Manage

Success

DEMONS

Give people

<https://app>

Build your app

App name *

DEMONS

Description *

Enter a summary

Describe your app.

App logo

Upload

Delete

App theme color

Global App Settings

Install this app automatically.

Hide app navigation pane.

Allow users to make a copy of the reports in this app.

Support site

Share where your users can find help

Contact Information

Show app publisher

Show items contacts from the workspace

Show specific individuals or groups

Enter a name or email address

Copy

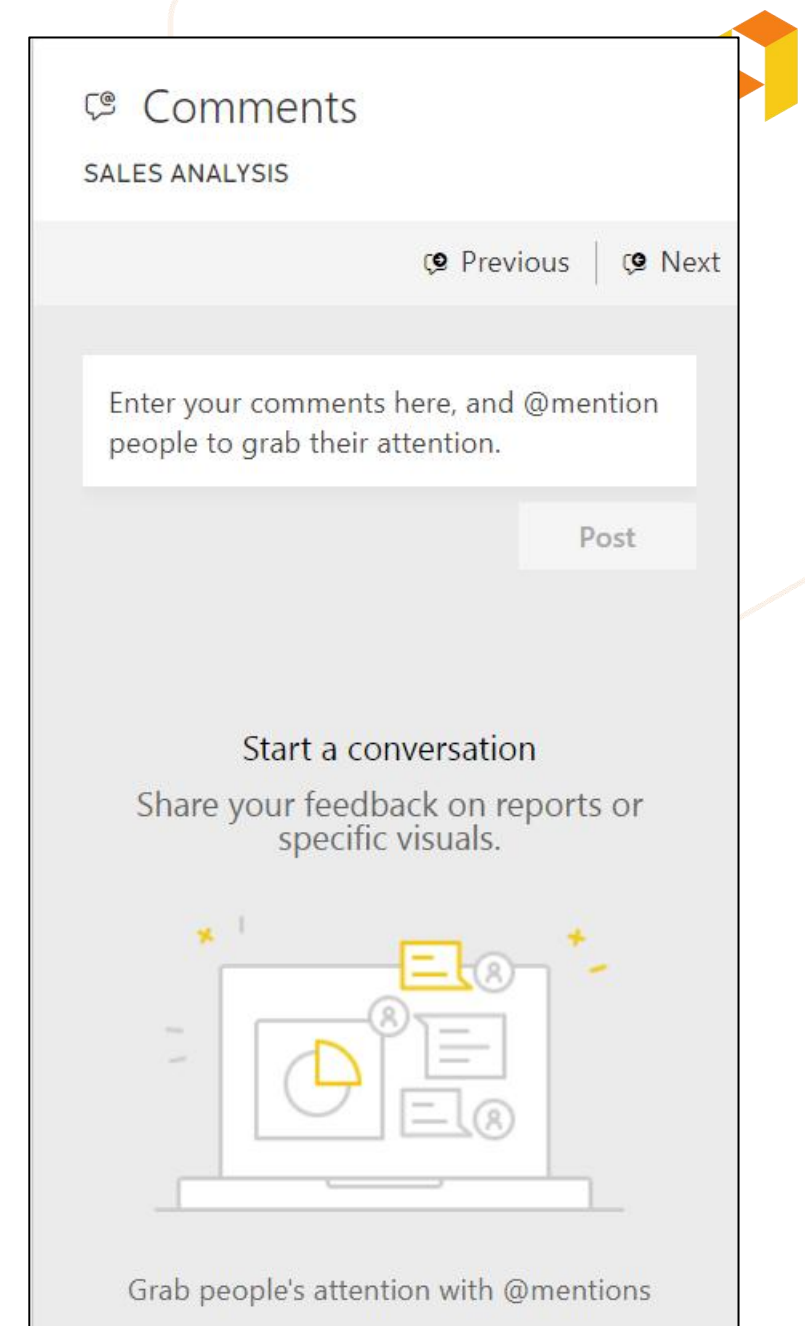
to app

Close

Workspace users

Provide feedback in report comments

- Comment pane
- Tag specific people
- Comment on specific visuals





Power BI licensing overview

License	Shared workspace	Premium workspace
Free	• Access to their own content. • Cannot distribute any content.	• Access workspace content. • Cannot distribute any content.
Pro	• Access to all workspace content. • Distribute to other Pro users.	• Access all workspace content. • Distribute content to all users.
Premium Per User (PPU)	• Access to all workspace content. • Distribute to other PPU users.	• Access all workspace content. • Distribute content to all users.

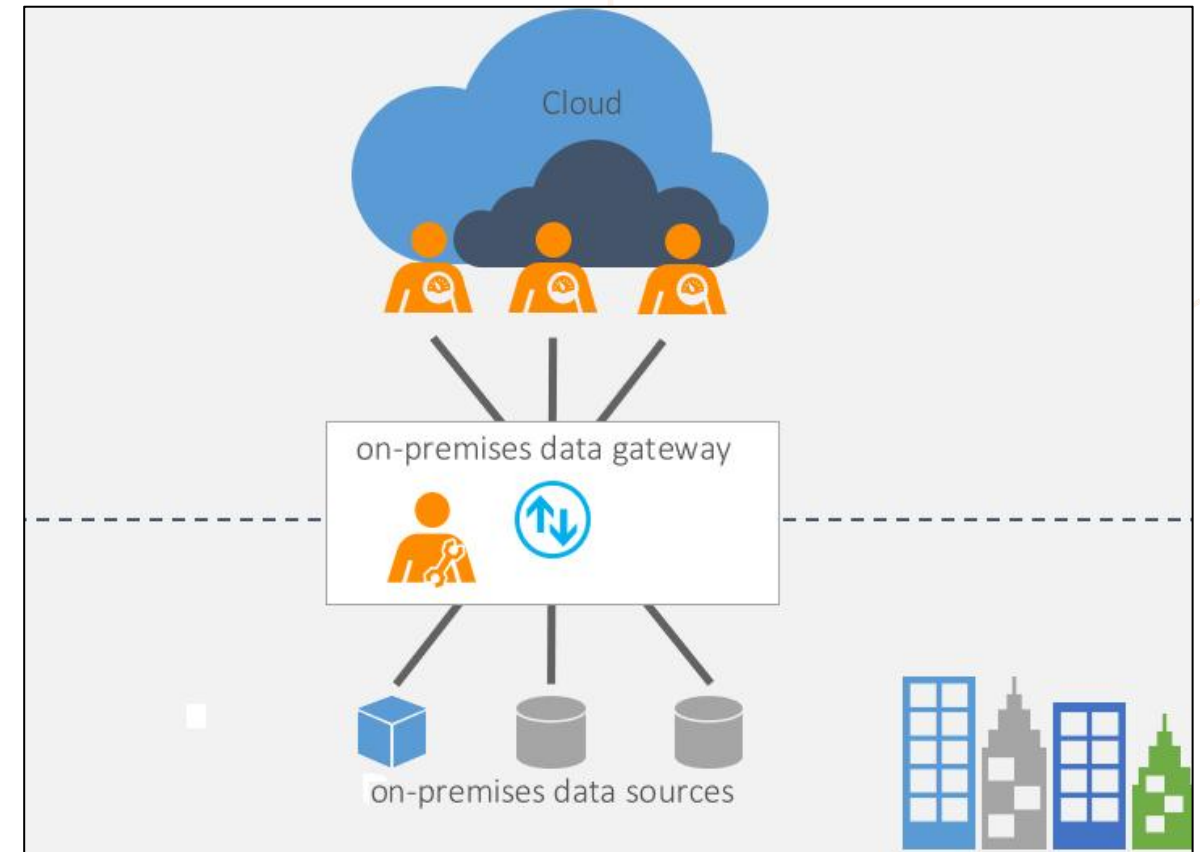
Manage semantic models





Connect to on-premises data sources

- Gateways allow Power BI to connect and refresh **on-premises data sources.**
- Organizational or personal gateways available.
- Require credentials to data source.





Keep data up to date

On-demand or scheduled refresh

- 8 times per day for shared
- Up to 48 times per day with Premium

Automatically pauses after

- 5 consecutive failures
- 2 months of inactivity

Scheduled refresh

Keep your data up to date

Configure a data refresh schedule to import data from the data source into the dataset.

☒ On

Refresh frequency

Daily ▼

Time zone

(UTC-08:00) Pacific Time (US and Can: ▼

Time

[Add another time](#)

Send refresh failure notifications to

☒ Dataset owner

☐ These contacts:

Enter email addresses

Apply

Discard



Incremental refresh

- Works best with query folding
- DirectQuery requires Premium/PPU

How to enable:

- Define the filter parameters
- Use the parameters to apply a filter
- Define the incremental refresh policy

Incremental refresh and real-time data

Refresh large tables faster with incremental refresh. Plus, get the latest data in real time with DirectQuery (Premium only). [Learn more](#)

 These settings will apply when you publish the dataset to the Power BI service. Once you do that, you won't be able to download it back to Power BI Desktop. [Learn more](#)

1. Select table

Orders 

2. Set import and refresh ranges

☒ Incrementally refresh this table

Archive data starting  before refresh date

Data imported from 1/1/2017 to 11/17/2022 (inclusive)

Incrementally refresh data starting  before refresh date

Data will be incrementally refreshed from 11/18/2022 to 11/20/2022 (inclusive)

3. Choose optional settings

☐ Get the latest data in real time with DirectQuery (Premium only) [Learn more](#)

Selected table cannot be folded for DirectQuery.

☐ Only refresh complete days [Learn more](#)

☐ Detect data changes [Learn more](#)

4. Review and apply

Archived

Incremental Refresh



Semantic model endorsement

- Provide trusted semantic models and encourage reusability.

Endorsement and discovery

Help coworkers find your quality content by endorsing this semantic model and making it discoverable. [Learn more](#)

☒ None
This semantic model will appear in search results but isn't endorsed.

☐ Promoted
When you're ready to distribute the semantic model to your coworkers, promote it to let them know.

☐ Certified
Certify your semantic model to show coworkers that it's been reviewed and meets your org's certification criteria. [How do I get my semantic model certified?](#)

☐ Make discoverable
Allow users without access to this semantic model to discover it and request permissions to access the data [Learn more](#)

Apply

Discard



Data protection report settings

- If and how users can export data.

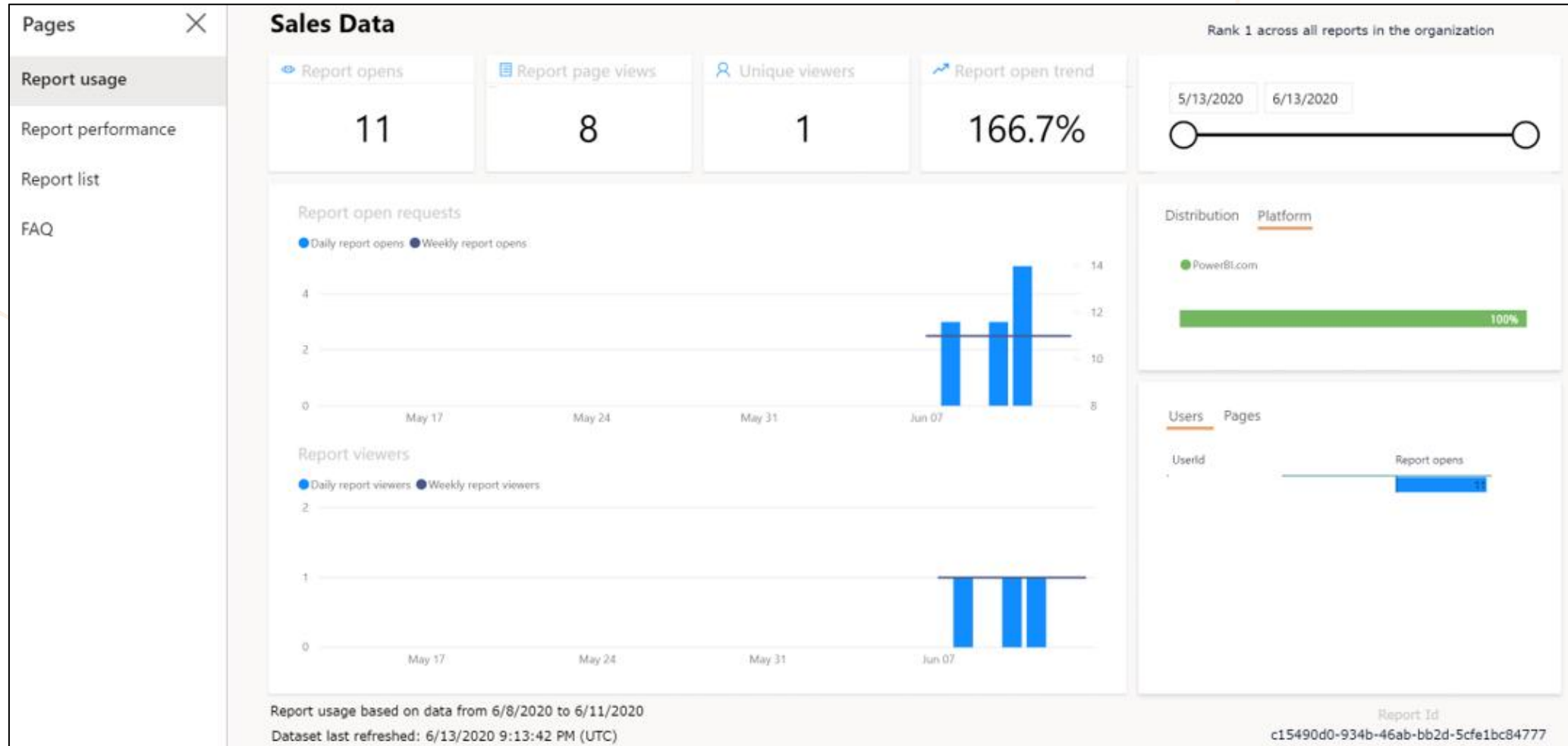
Export data
Choose the type of data you allow your end users to export.
Summarized data and data with current layout
Summarized data and data with current layout
Summarized data, data with current layout and underlying data
None

- Set data sensitivity.

Sensitivity label *
Classify the sensitivity of this report content.
Non-Business
Non-Business
Public
General



Leverage Usage Metrics report for insights



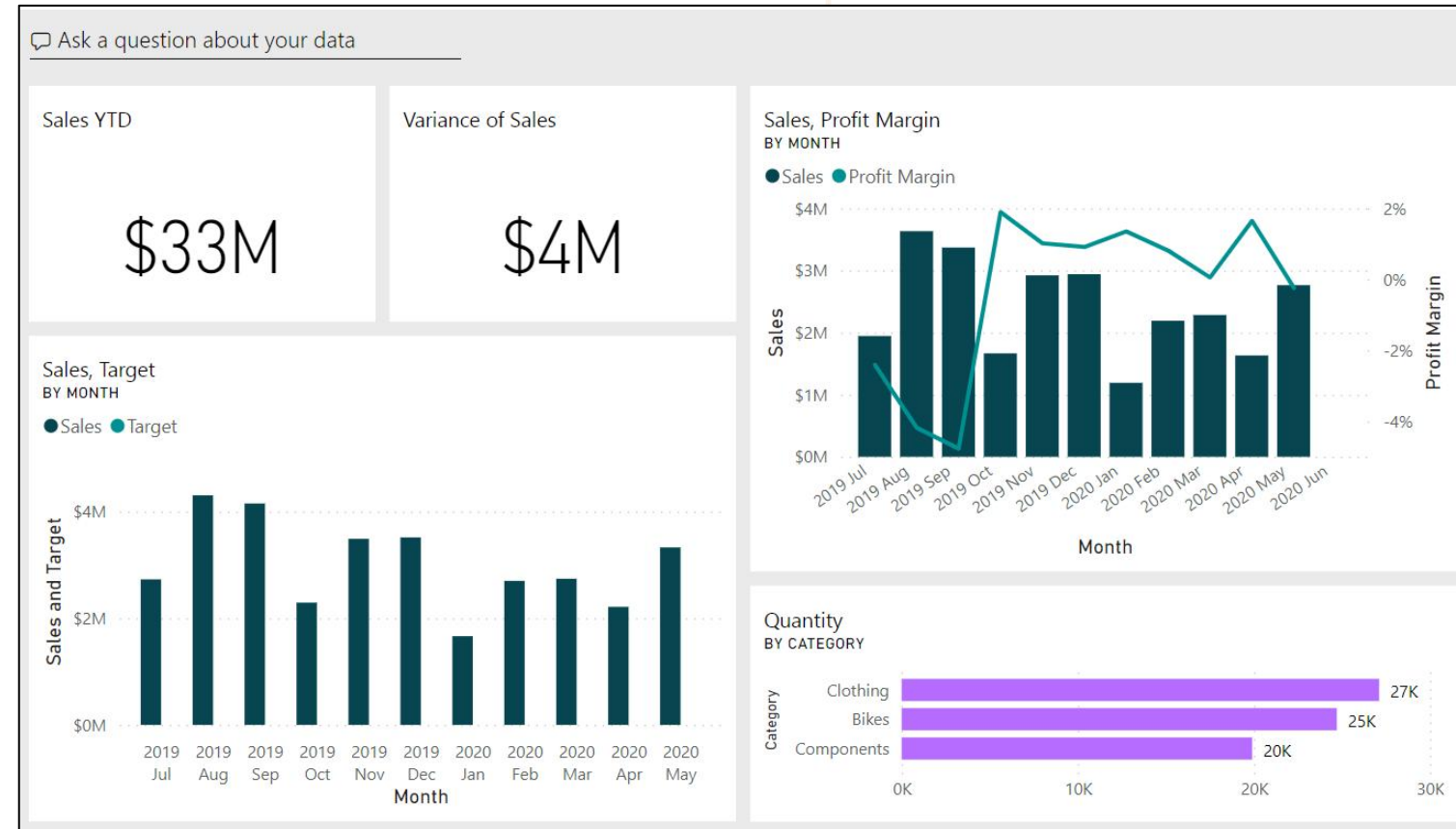
Create dashboards





What is a Power BI dashboard?

- Single page
- Visuals pinned from reports
- Tiles are static
- Default Q&A feature
- Refreshes with semantic model
 - DirectQuery every 15 min





Pin visuals to a dashboard

- Specific visuals
- Entire pages (pin live)

Pin to dashboard

Select an existing dashboard or create a new one.

Where would you like to pin to?

☒ Existing dashboard

☐ New dashboard

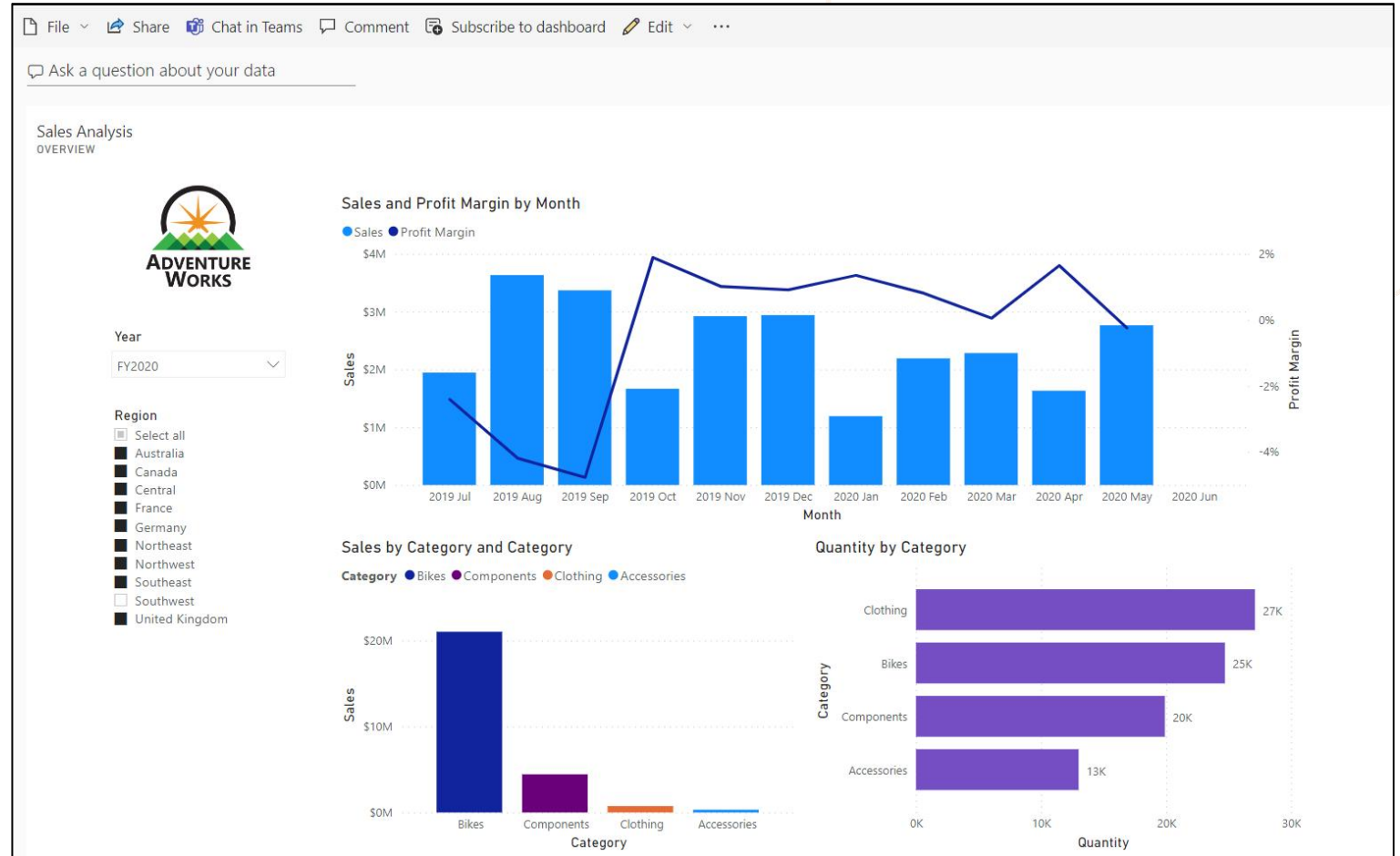
Select existing dashboard

Sales Overview

Pin live page enables changes to reports to appear in the dashboard tile when the page is refreshed.

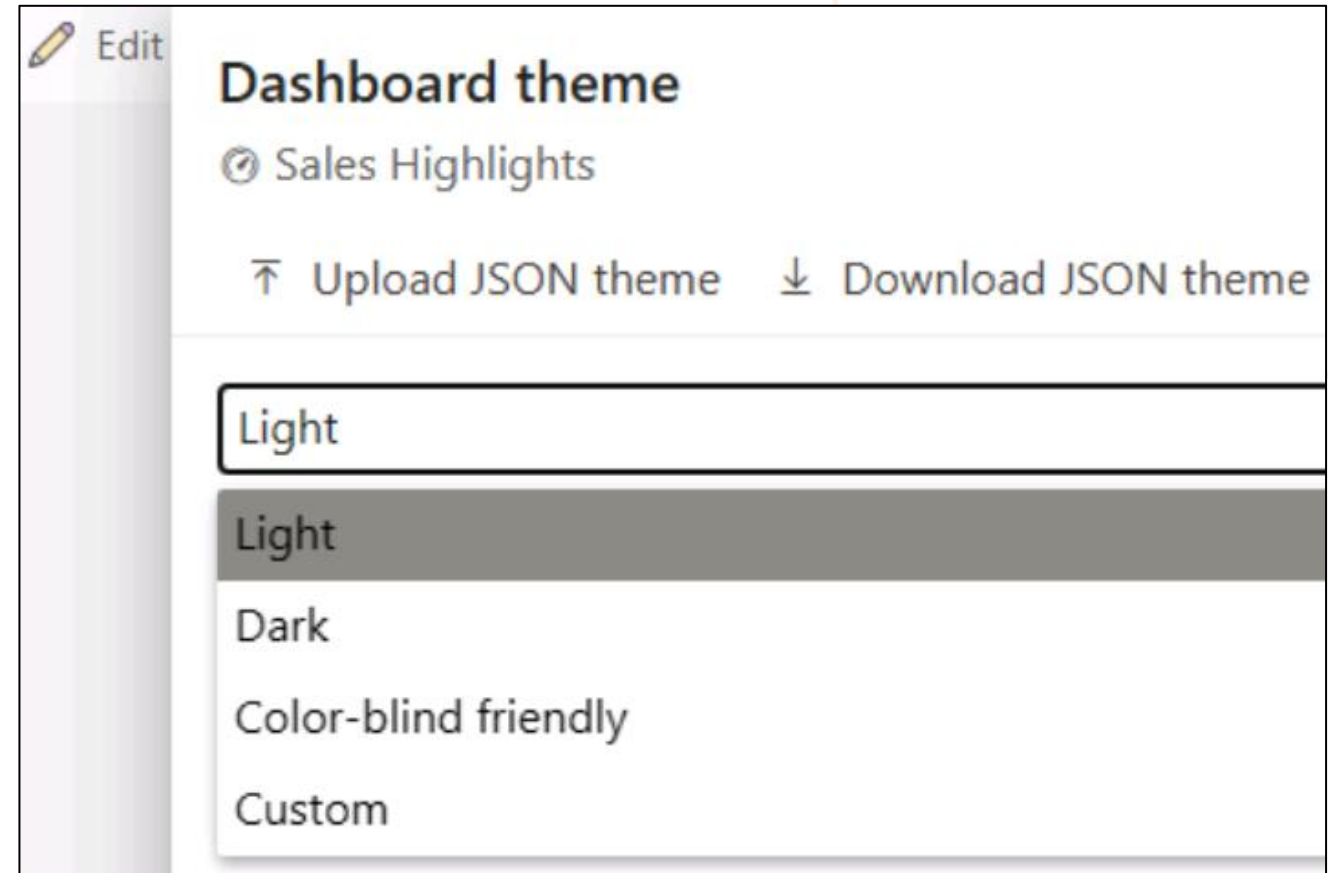
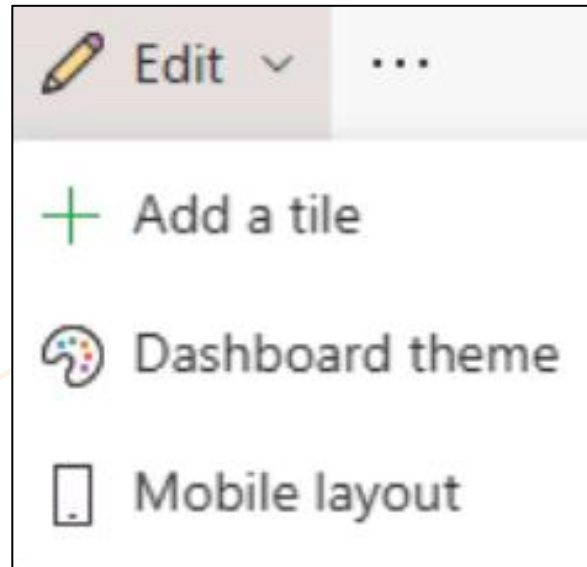
Pin live

Cancel



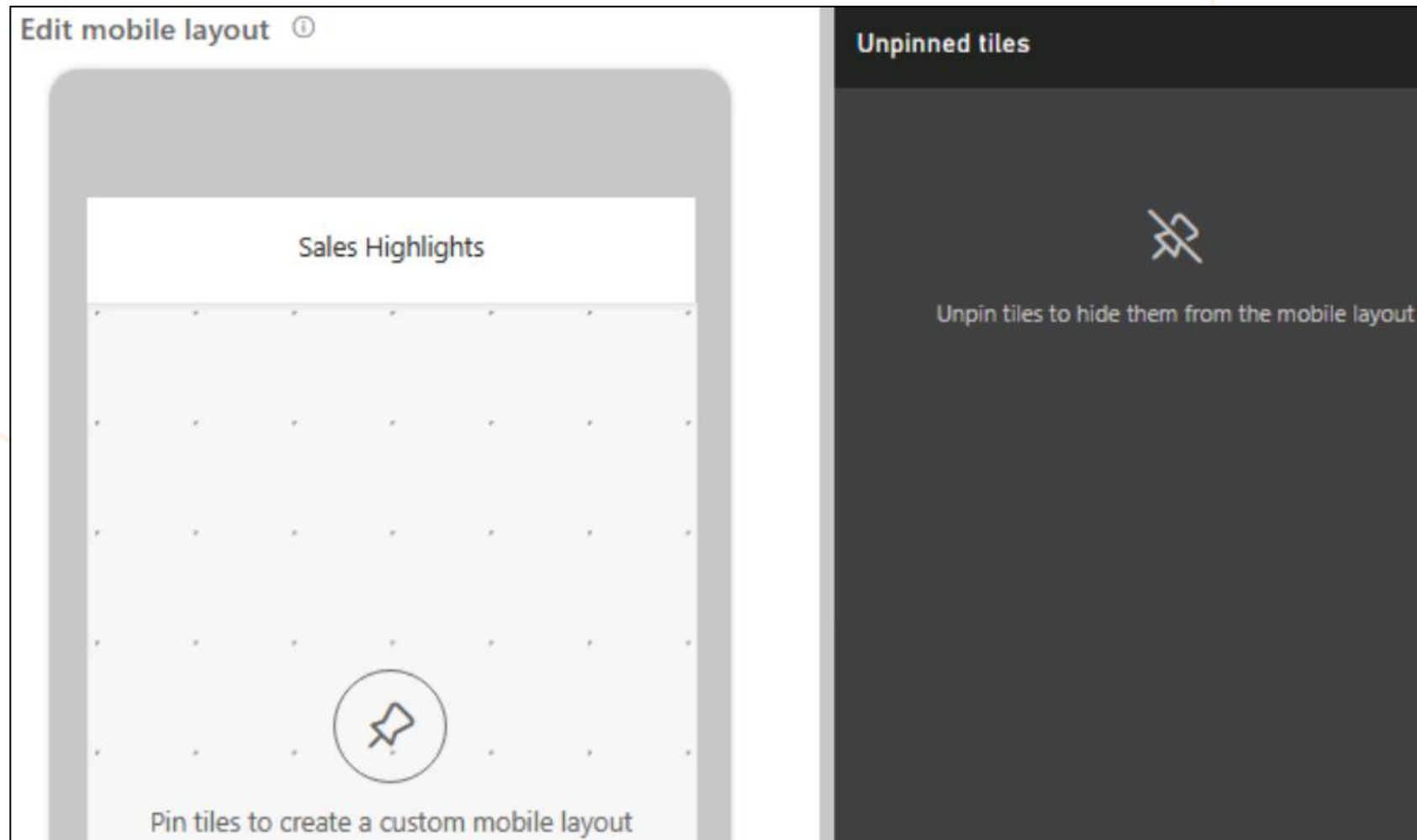


Enhance your dashboard with a theme



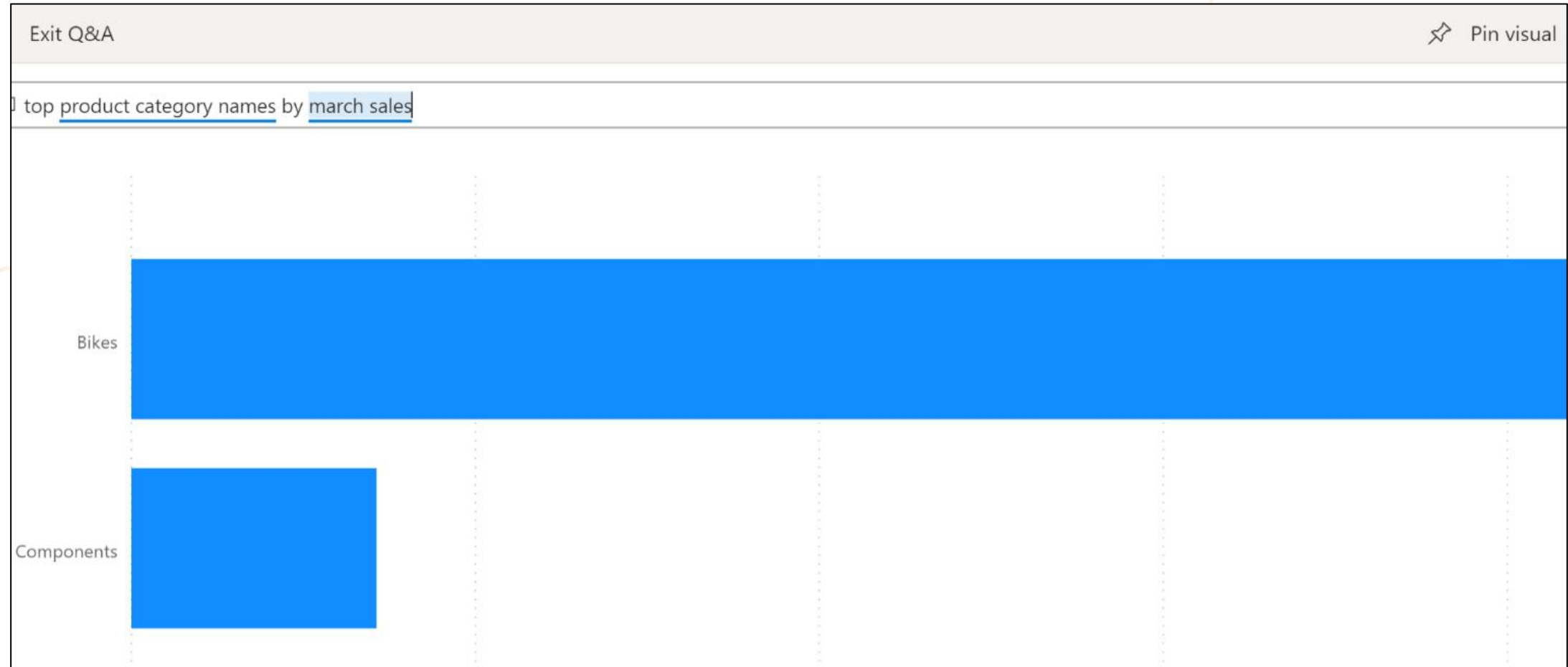


Design mobile layout for dashboards on the go





Use natural language to explore with Q&A





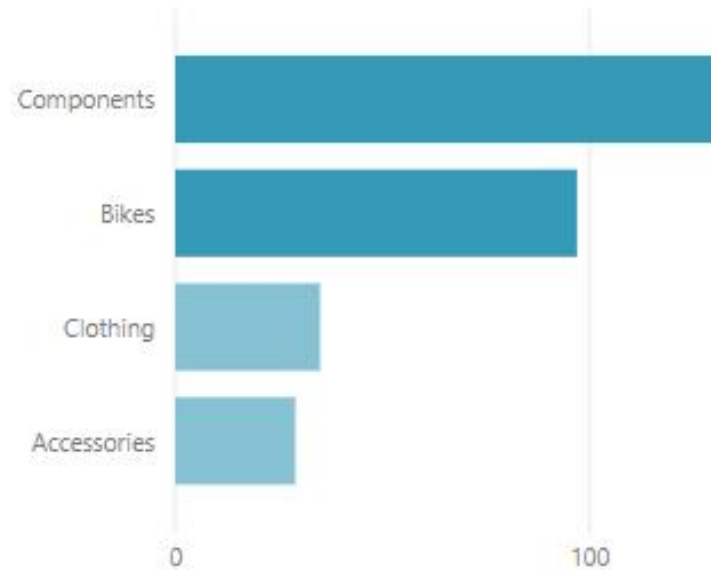
Discover more with Quick Insights

- Uses Machine Learning algorithms.
- Generate visual interactions.
- Find insights missed during development.

Quick Insights

A subset of your data was analyzed and the following insights were found. [Learn more](#)

Count of Product
BY CATEGORY NAME



'Components' and 'Bikes' have noticeably more Product.

Explore Copilot for Power BI





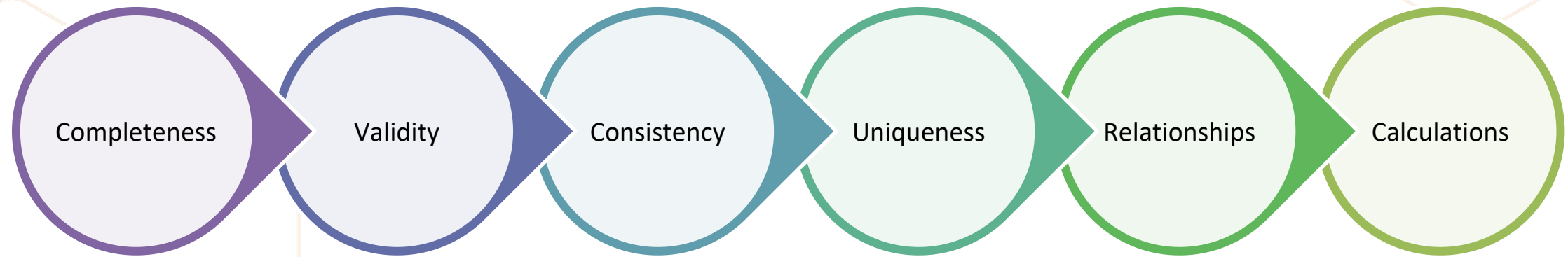
How to use Copilot for Power BI

- Create measures based on natural language.
- Update the semantic model with synonyms for improved user Q&A experience.
- Generate report content, summary visuals, and pages from prepopulated prompts.
- Analyze a summary of underlying semantic model.



Prepare your data and ensure quality

Copilot isn't effective without good data modeling.





Query with DAX

```
// DAX query generated by Fabric Copilot with "total sales for all salespeople individually for all items in the accessories category"
// Filter for Accessories category
EVALUATE
    CALCULATETABLE(
        SUMMARIZECOLUMNS(
            'Salesperson'[Salesperson],
            "Total Sales", SUM('Sales'[Sales])
        ),
        'Product'[Category] = "Accessories"
    )
```

 total sales for all salespeople individually for all items in the accessories category

Explain this query

Explain a DAX topic

Retry

Keep query

Discard query

Content created by AI may contain mistakes, so review it carefully. [Read terms](#)

Salesperson	Total Sales
Stephen Jiang	8374.76
Michael Blythe	38682.84
Linda Mitchell	66916.05

Create measures

- Suggest new measures
- Evaluate returns results
- Update model: Add new measure

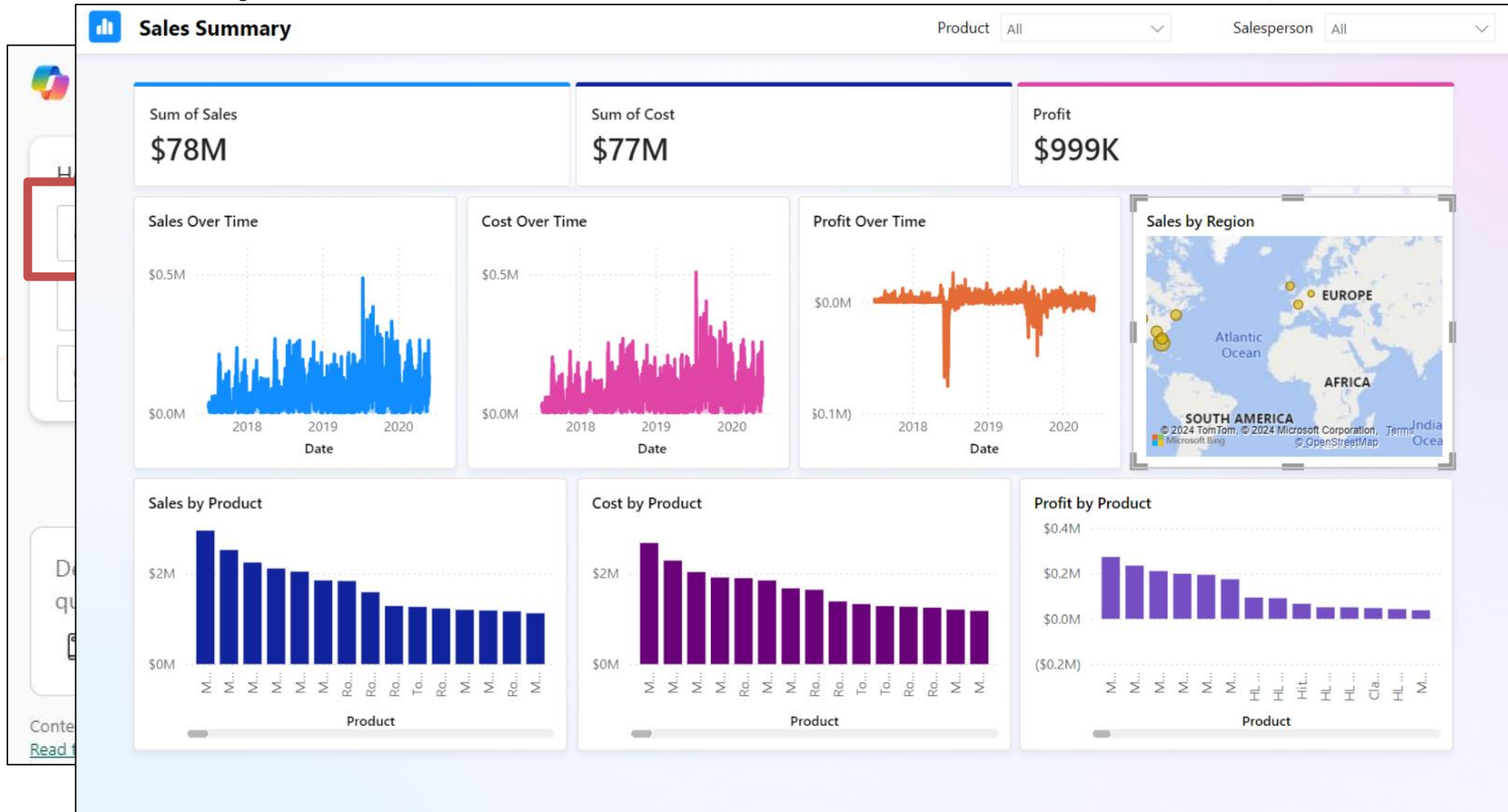
```
1 // DAX query generated by Fabric Copilot with "Suggest new measures in a DAX query for further
2 analysis and try them out with one or more suitable columns"
3 DEFINE
4 // New measure to calculate the average profit per product sold
5 Update model: Add new measure
6 MEASURE 'Sales'[Avg Profit per Product] = DIVIDE([Profit], [Unique Products Sold])
7 // New measure to calculate the average sales per reseller
8 Update model: Add new measure
9 MEASURE 'Sales'[Avg Sales per Reseller] = DIVIDE([Total Sales], [Unique Resellers])
10 // New measure to calculate the average quantity per order
11 Update model: Add new measure
12 MEASURE 'Sales'[Avg Quantity per Order] = DIVIDE([Total Quantity], [Orders])
13 // New measure to calculate the average sales per order
14 Update model: Add new measure
15 MEASURE 'Sales'[Avg Sales per Order] = DIVIDE([Total Sales], [Orders])
16
17 // Evaluate the new measures
18 EVALUATE
19 ROW(
20     "Avg Profit per Product", [Avg Profit per Product],
21     "Avg Sales per Reseller", [Avg Sales per Reseller],
22     "Avg Quantity per Order", [Avg Quantity per Order],
23     "Avg Sales per Order", [Avg Sales per Order]
24 )
```

Results | Result 1 of 1 ▾ | Copy ▾

	[Avg Profit per Product]	[Avg Sales per Reseller]	[Avg Quantity per Order]	[Avg Sales per Order]
1	2992.5	122703.43	56.45	21445.95



Create reports





Suggested and custom prompts

Product Cost and Profitability Analysis

Subcategory: All Category: All

Profit

\$999K

Profit Margin

1.29%

Profit by Product Category

Category	Profit
Components	\$1M
Clothing	\$0.5M
Accessories	\$0.5M
Bikes	\$0.5M

Profit Margin by Product Subcategory

Subcategory	Profit Margin
Tires and Tubes	37.43%
Socks	37.12%
Hydration Packs	36.79%
Cleaners	36.53%
Bottles and Cages	36.36%
Bike Racks	36.26%
Vests	35.61%

Profit Trend Over Time

Accessories Bikes Clothing Components

Profit Margin Trend Over Time

Bib-Shorts Bike Racks Bottles an... Bottom Br... Brakes Caps Chains Cleaners

OK, here's a suggested outline for your report. Select any page topic to view details and start creating pages.

Sales Performance by Salesperson

Regional Sales Analysis

Product Cost and Profitability

Create a page to analyze the cost and profitability of different products, including standard cost, profit, and profit margin.

+ Create

Reseller Performance

Is this response helpful?

Create a page to analyze the cost and profitability of different products, including standard cost, profit, and profit margin.

Created a Product Cost and Profitability Analysis page.

Copyright Nawatech
2025

Generate summaries

Create a narrative with Copilot



Describe the summary you want and the visuals it should reference. AI-generated content can contain mistakes, so review your summary carefully. [Read terms](#)

Give an executive summary

Answer likely questions from leadership

Create a bulleted list of insights

0/1000

Reference visuals

Current page

Update

☒ Select all

☒ Sales Summary (Current)

☒ Card (new)

☒ Sales Over Time



Generate summaries

Sales Summary:

Sales by Region:

- The **Southwest region** leads in sales with **\$18M**.
- **Canada** has the second-highest sales at **\$13.88M**.
- **Germany** and **Australia** registered the lowest sales figures at **\$1.88M** and **\$1.39M** respectively ¹.

Sales by Product:

- The **"Mountain-200 Black, 38"** model records the highest sales, totalling **\$2.94M**.
- Significant product sales also include the **"Mountain-200 Black, 42"** and **"Mountain-200 Silver, 38"** models, at **\$2.52M** and **\$2.24M** respectively.
- Lower sales products include the **"LL Road Frame - Red, 52"** and **"Short-Sleeve Classic Jersey, S,"** with sales of **\$20.1K** and **\$45.66K** respectively ².

Sales Over Time:

- A significant spike in sales occurred on **2019-07-15**, totalling **\$486.2K**.
- Another notable sales peak was on **2018-09-13**, with sales reaching **\$262.54K**.
- Sales on **2018-11-27** were lower at **\$136.84K** ³.

Profit:

- The overall **profit** was recorded at **\$999.49K**, with total sales amounting to **\$77.55M** and costs at **\$76.55M** ⁴.

Profit by Product:

- The **"Mountain-200 Black, 38"** generated the highest profit at **\$272.77K**.
- Products such as **"HL Mountain Frame - Silver, 38"** and **"Classic Vest, S"** also recorded notable profits, at **\$93.74K** and **\$47.31K** respectively.
- Products like the **"HL Touring Frame - Blue, 50"** and **"Touring-3000 Blue, 62"** registered losses at **(\$929.15K)** and **(\$9.03K)** ⁵.

Profit Over Time:

- The highest profit on a single day was recorded on **2017-08-01**, at **\$3.65K**.

Created with AI. Inaccuracies are possible. [Read terms](#)



Thank you
Let's discuss our collaborations

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